

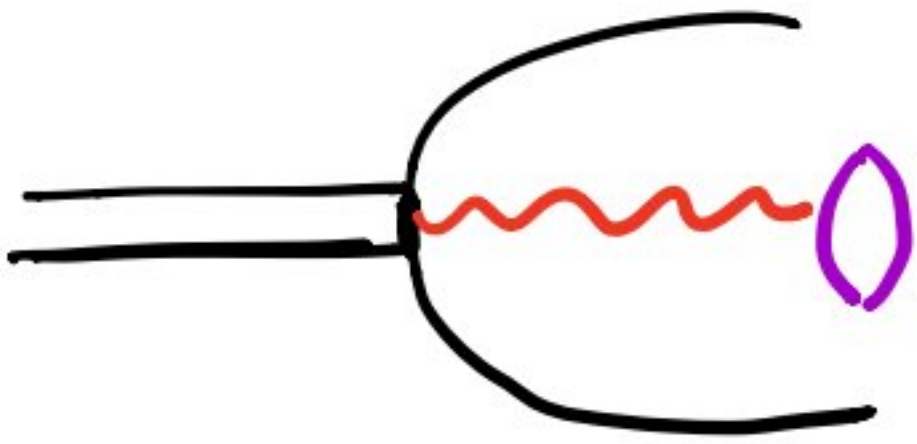
VITREOUS:

- Vitreous humour → • Transparent gel
 - 99% H_2O
 - Hyaluronic acid → give the gel consistency
- ❌ • TYPE II COLLAGEN

Embryology:

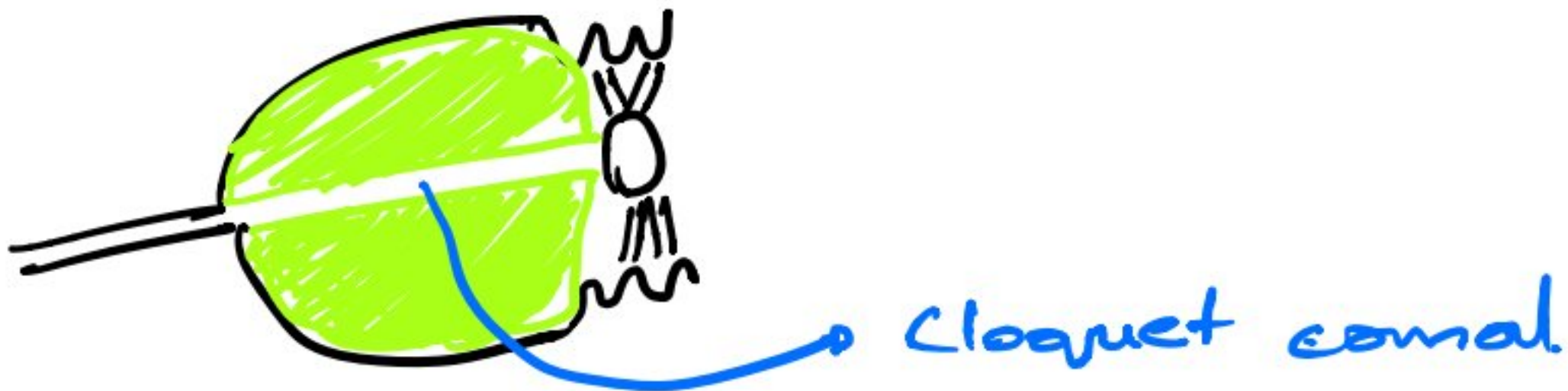
- 1° → Primary vitreous → Hyaloid artery (only in fetus) ← Mesoderm
- 2° → Secondary " → Vitreous gel } Neuroectoderm
- 3° → Tertiary " → Zonules

Hyaloid artery: Supplies nutrition to lens (in fetus)



Regression of 1° vitreous & its remnants

1) Complete Regression



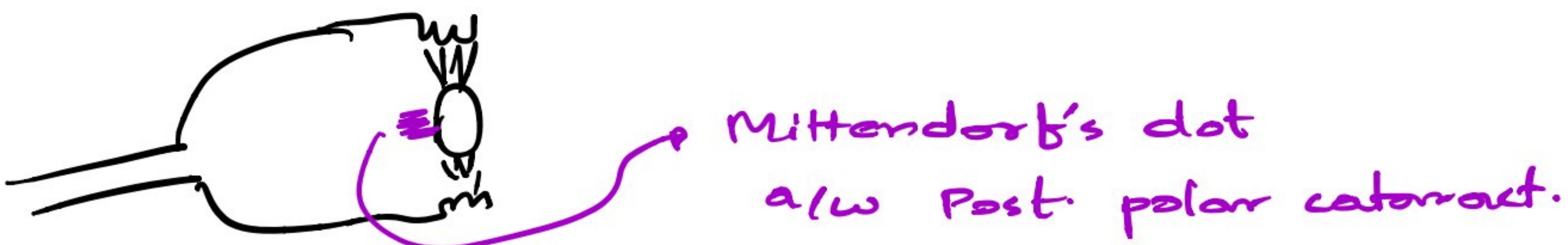
2) Incomplete regression



3) Persistants of attachment to optic disc

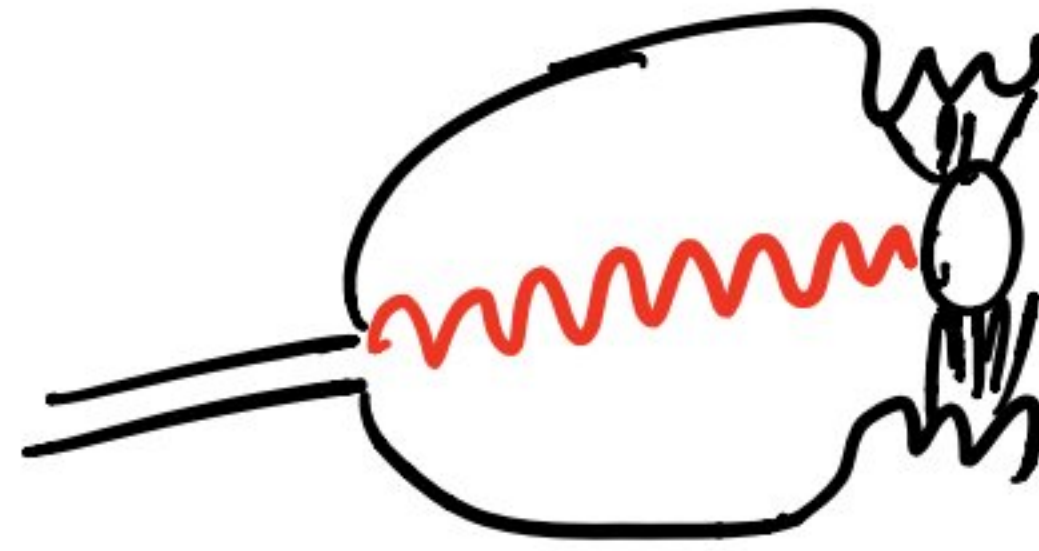


4) Persistants of attachment at P surface lens



PATHOLOGICAL → PHPV : Persistent Hyperplastic 1° Vitreous → a/w ^{*}Patau Syndrome (Trisomy 13)

Symptoms → i) Leukocoria
ii) Microphthalmos
a/w Congenital Cataract



PHPV



preladder CI

Investigation: USG B-Scan



Complication:

- 1) Intra Op: Vitreous H'mage during congenital cataract sx
- 2) Tractional RD & d/t fibrous band

VITREOUS DEGENERATION

^{*}1) Asteroid Hyalosis → Calcific degeneration

- moves with eye movement return to same position
- No vision loss
- a/w DM / HTN / Hyperlipidemia
- Old age ♂
- Not a/w myopia

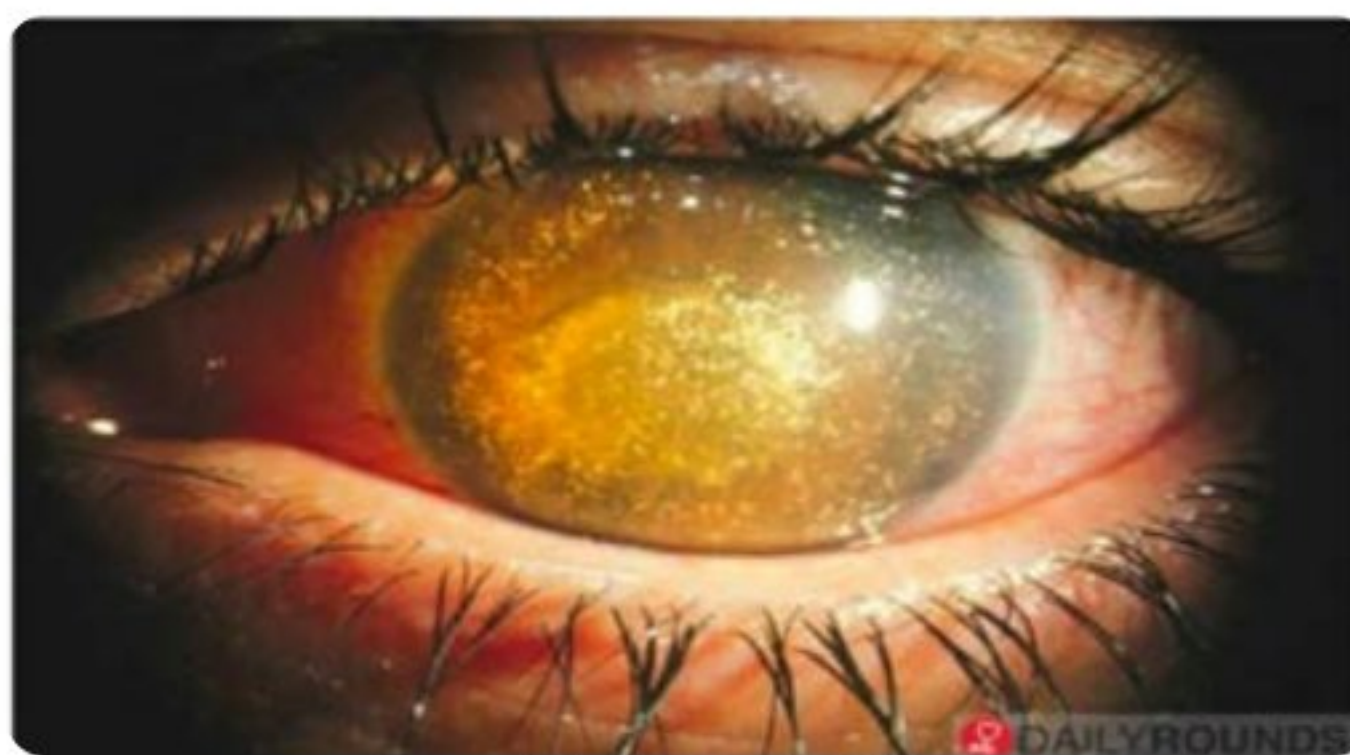
ASTEROID HYALOSIS



^{*}2) Synchysis Scintillans → Lipid degeneration [cholesterol crystals]
(Golden Shower)

SYNCHYSIS SCINTILLANS

- move i eye movement & don't return the same position → sink down
- Cholesterol come → Vitreous H'mage (RBC cell memb)



Synchysis Scintillans

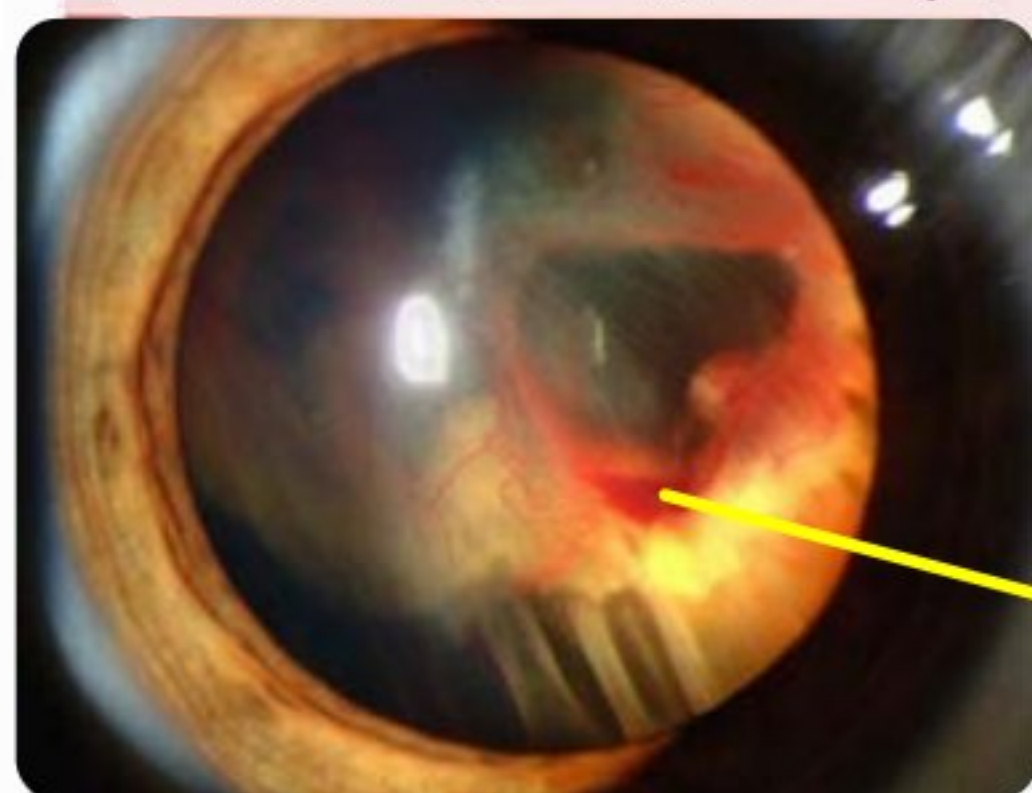
SYNCHESIS → Liquification of vitreous
SYNERESIS → Collapse of vitreous

Vitreous H'mage

Presentation:

- ① Blurred vision
- ② Floaters → faint cobweb like apparitions floating through the field of vision.
- ③ Reddish tint of vision
- ④ Photopsia → brief flashes of light in the peripheral vision.

VITREOUS H'MAGE



Boat shaped H'mage

Metamorphopsia won't be seen in vitreous H'mage

Causes of V.H:

- 1) PDR → MC in Adults
- 2) Trauma → MC in children
- 3) Eel's d/s → Recurrent Vitreous H'mage in an apparently healthy young adult ♂
- 4) ROP
- 5) Sickle cell Retinopathy
- 6) COAT'S D/s
- 7) CRVO

Management: Observation for 6 months
for T2DM → Observe x 3 months
T1DM → Observe x 1 month

↓ does not resolve

Pars plana Vitrectomy.

Complication of VH:

- Lysed RBC ⇒ Khaki coloured cell
- ↓
- 2° OAG / Khaki - cell glaucoma.

Intravitreal injections: 26 G needle → Injected in pars plana pointed to mid-vitreous chamber.

- 1) Phakic eye : 4.5mm from limbus
- 2) Pseudophakic eye : 4.0mm from limbus
- 3) Aphakic eye : 3.5mm from limbus

Complication intravitreal injection

1) (MC) → ↑ IOP
L/E CRAO } Rx → Paracentesis of Aqueous humor

always dilate for Fundus Examination to rule out impending CRAO
§ Arterial pulsation @ disc §

- 2) Cataract (PSC)
- 3) Endophthalmitis
- 4) Latrogenic Rhegmatogenous RD

Substance:

ANTIBIOTICS

- 1) Vancomycin
- 2) 3rd gen Cephalosporin
- 3) Amikacin

Gentamycin → Retinal Toxicity

STERIODS

- 1) Triamcinolone
- 2) Dexamethanone
- 3) Fluocinolone.

ANTI FUNGAL

- 1) Amphotericin B
- 2) Voriconazole

ANTI VEGF

- 1) Bevacizumab
- 2) Ranibizumab
- 3) Pegaptamib

ANTI VIRAL

Gancyclovir

ANTI CANCER DRUGS

- 1) MTx → 1^o intraocular lymphoma
- 2) Melphalan } Retinoblastoma
- Topotecan }

Glaucoma →
• C/c progressive optic neuropathy
• multifactorial
• occur d/t death of retinal ganglion cells
resulting in:
i) Characteristic optic disc changes
ii) " " visual field defect

IOP → Most significant modifiable risk for Glaucoma.

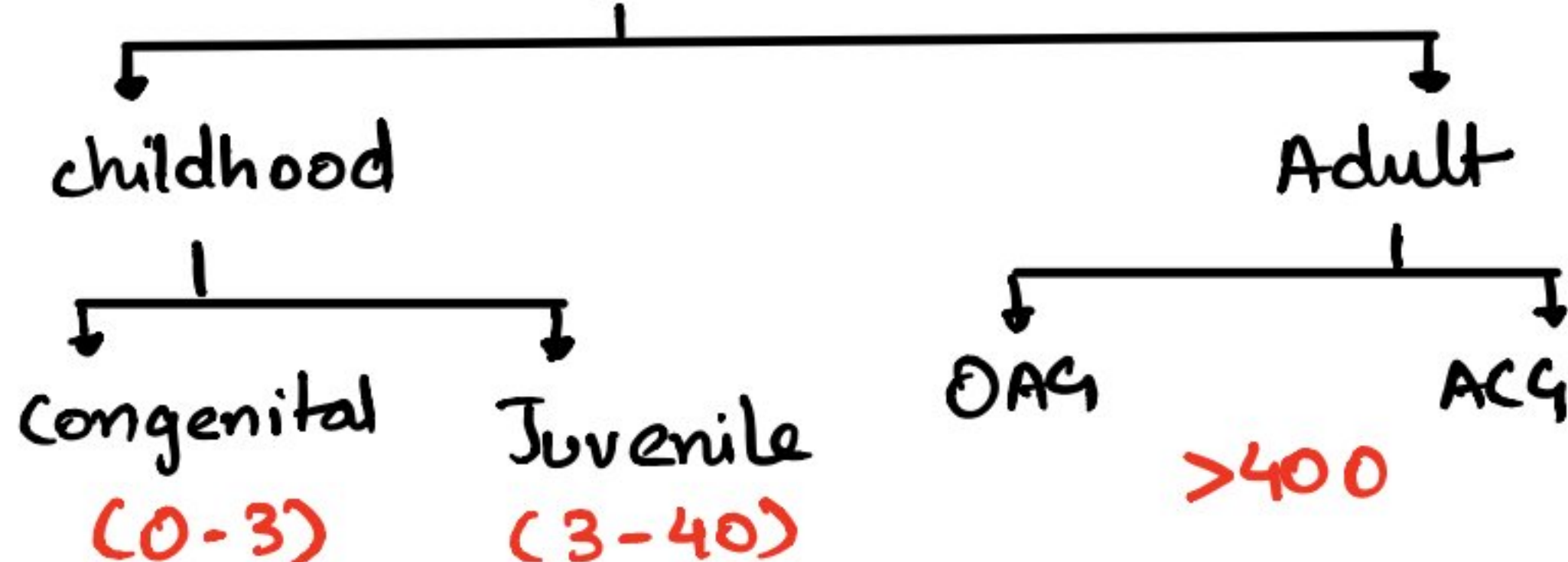
q: (MC) Aqueous production rate → 2-3 μL/min

Primary

Secondary

• Reason unknown

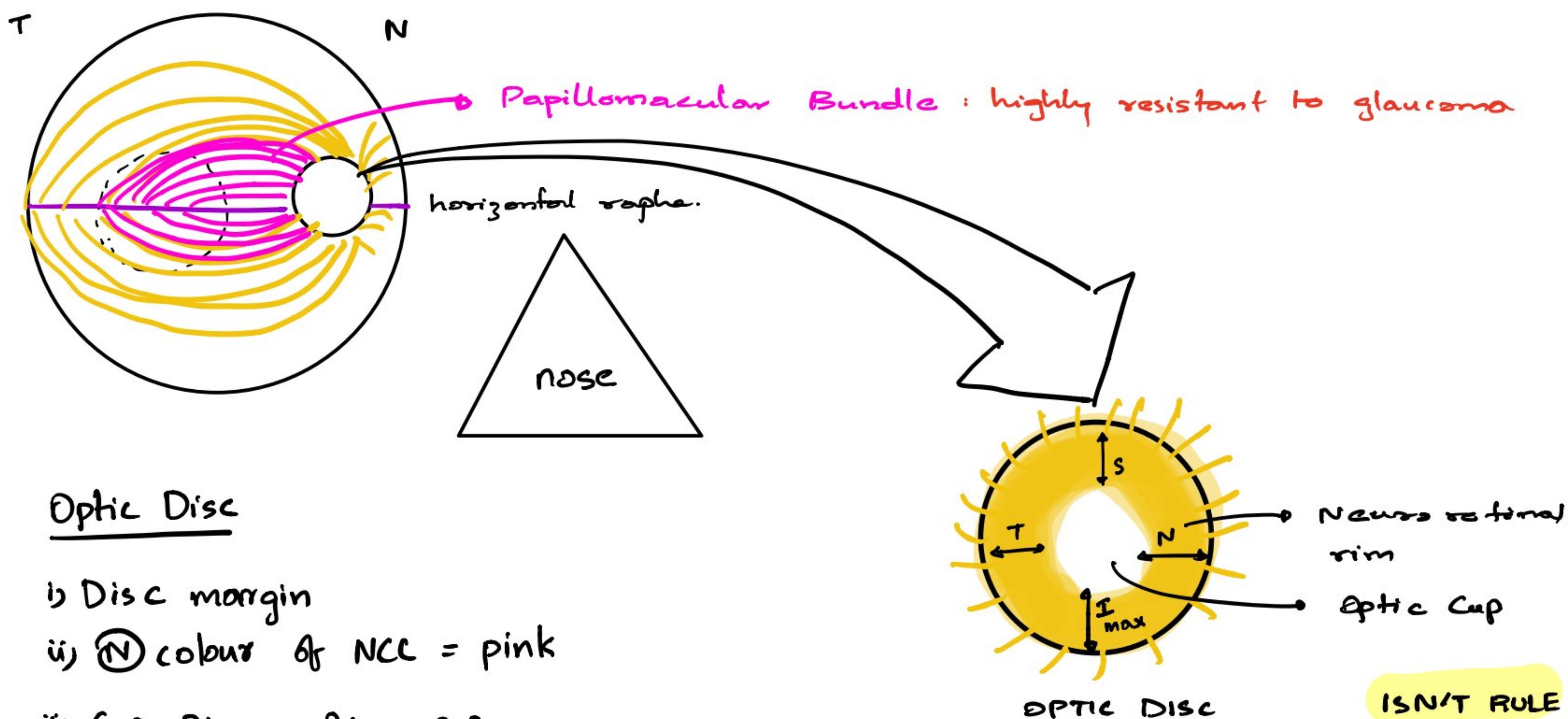
classification



- Uveitis (Especially anterior)
- Neovascular
- Lens induced
- Traumatic

↓
d/t → anterior
synechiae
steroid induced
trabecular mesh
work
blockage

Optic Disc / Papilla / Optic nerve head :



Optic Disc

i) Disc margin

ii) (N) colour of NCC = pink

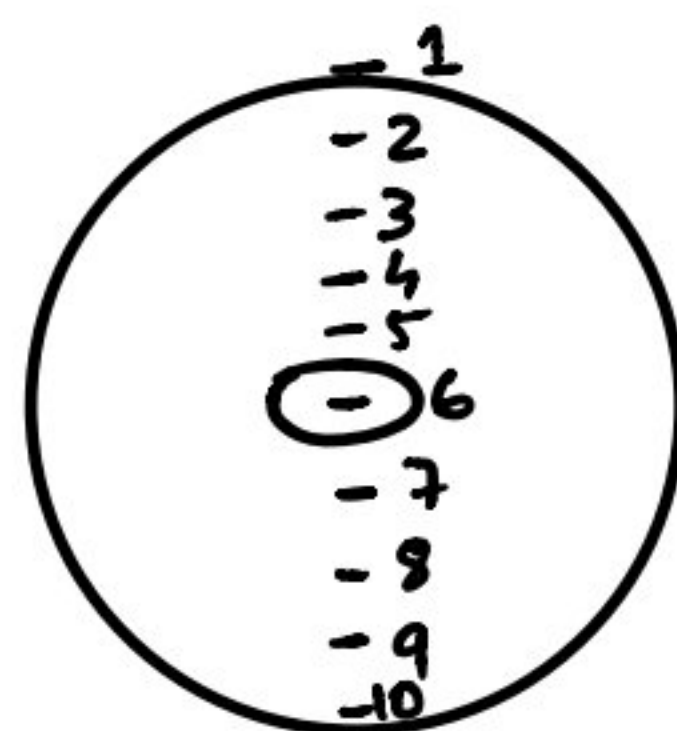
iii) Cup: Disc = $2/10 = 0.2$

Absolute = (N) upto 0.6

Relative = difference upto 0.2 (N)

(R) CDR = 0.5

(L) CDR = 0.3



ISN/T RULE
 $I_{max} > S > N > T$
Thickness

Optic Atrophy: When all optic N. fibres are damaged

1) $1^{\circ} \Rightarrow$ No optic disc edema $\rightarrow$ white margin

2) $2^{\circ} \Rightarrow$ Conditions causing disc edema $\rightarrow$ Blurred or Grey Margin

3) Glaucoma

4) Consecutive $\Rightarrow 2^{\circ}$ Retinal pathology

$\rightarrow$ Trauma { Retrobulbar }
Tumour

Toxin $\rightarrow$ Tobacco / Chloroquine / Alcohol $\rightarrow$ Ethanol
Methanol

ATT $\rightarrow$ i) INH

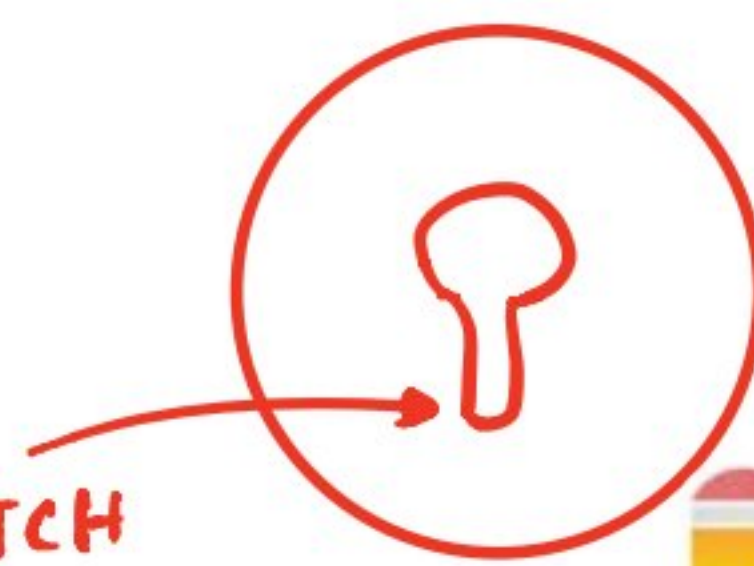
ii) Chloroquine (Not Rifampicin)

iii) Streptomycin

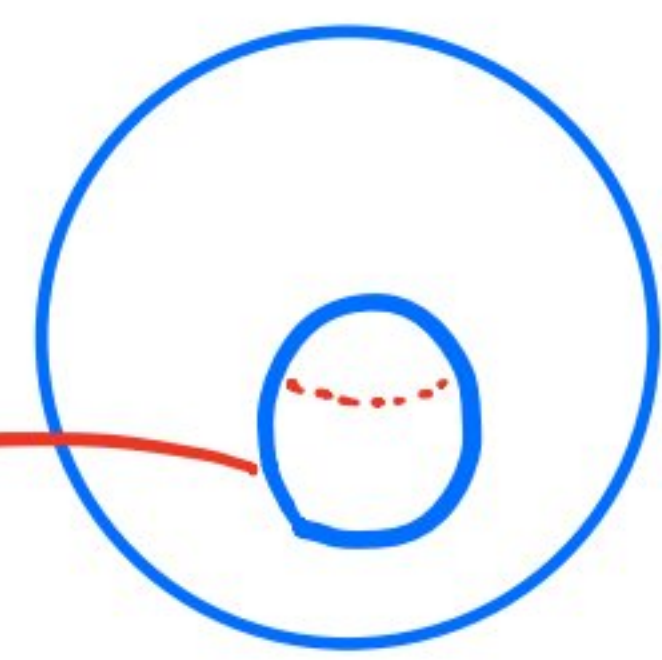
Glaucomatous Optic Disc changes

• Inferior retinal ganglion cells are affected first $\rightarrow$ ISN/T RULE broken

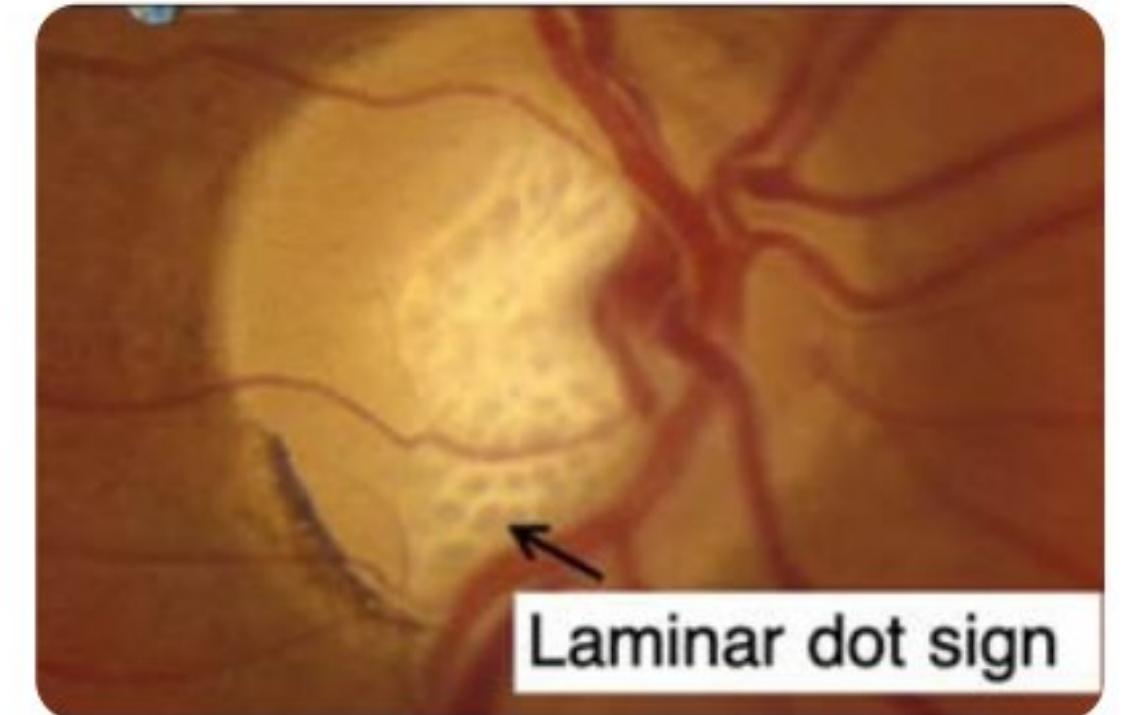
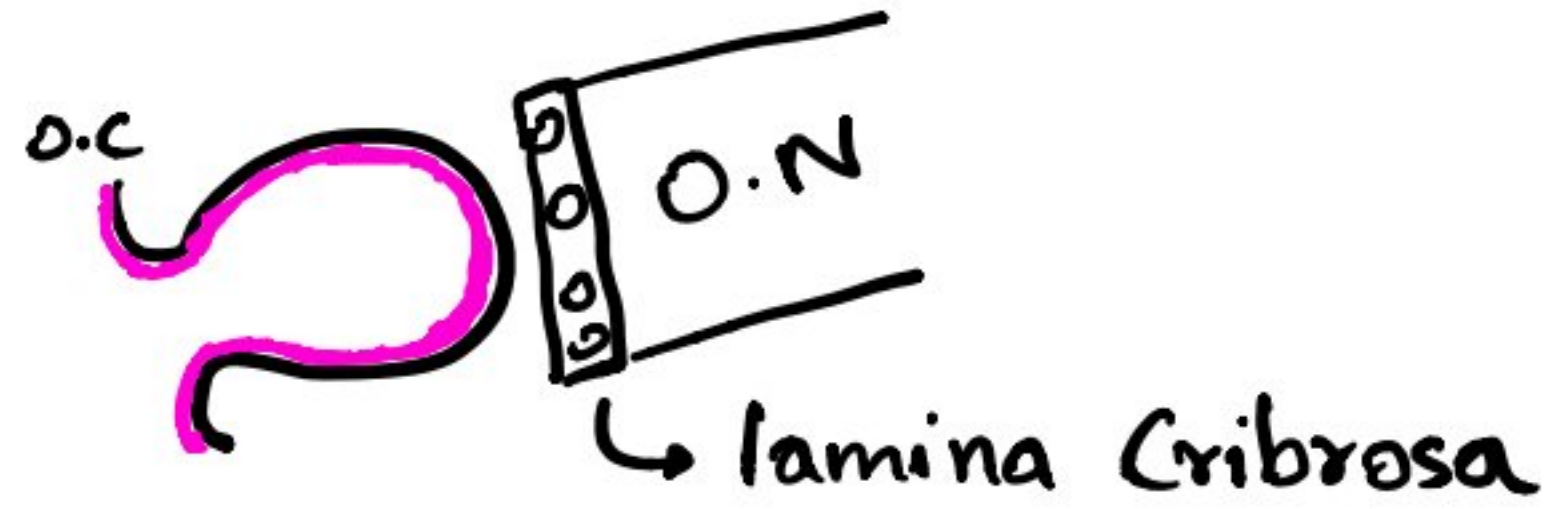
localised loss of NPR : RIM NOTCH



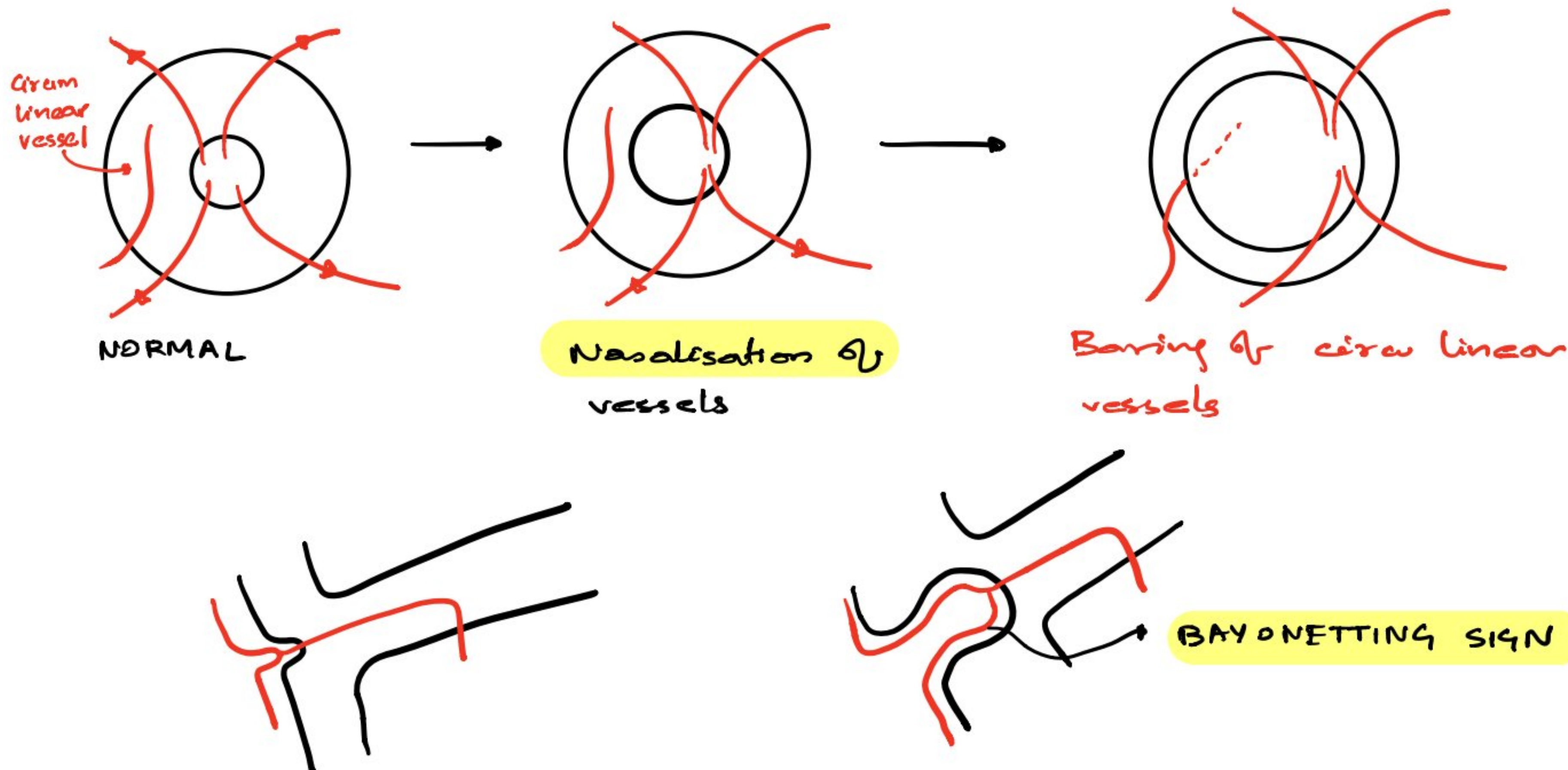
Generalised loss of NRR → Vertical elongation of cup



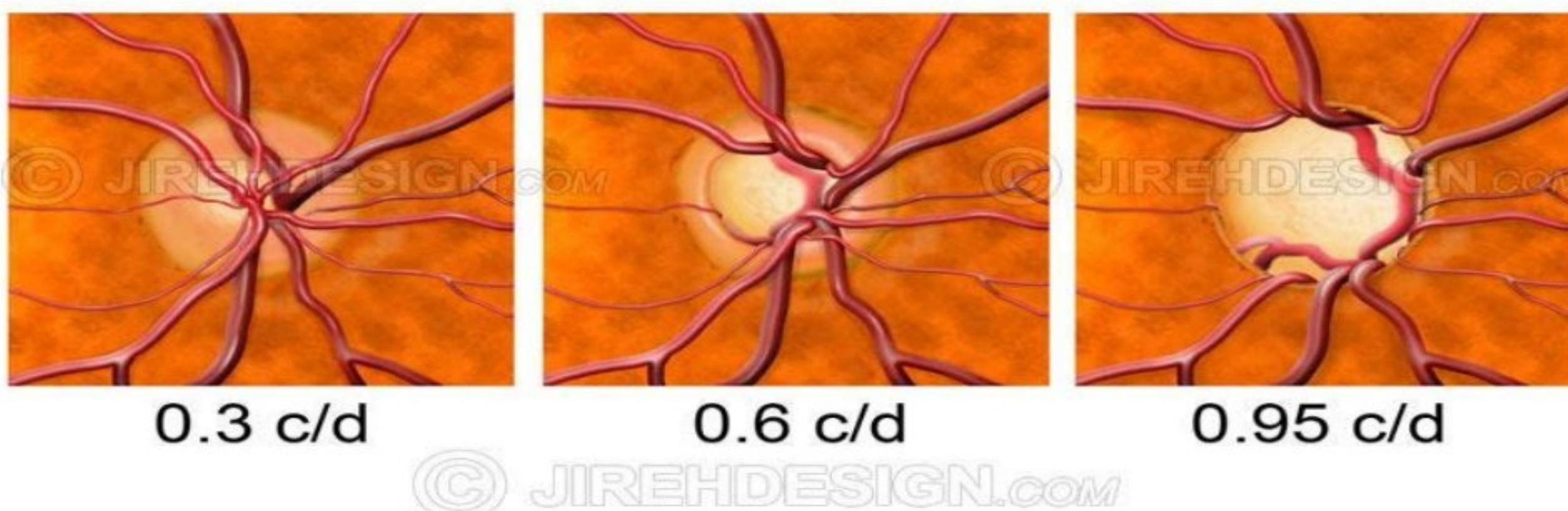
- Enlargement of cup = Thinning of all rims
- ↑ CDR
- Cupping: deepening of cup
 - ↳ "BEANPOT" Cupping



- Optic atrophy (1°) / Cavernous optic atrophy



Optic nerve head cupping progression



Healthy eye with a normal optic nerve head

Glaucoma eye with glaucomatous cupping of the the nerve head

Optic Disc Assessment:

i) Direct Ophthalmoscope

ii) IDO

iii) Slit lamp (Best method) → using concave lens (CHRUBY LENS) = -58.6 D attached to slit lamp
↓
using convex lens

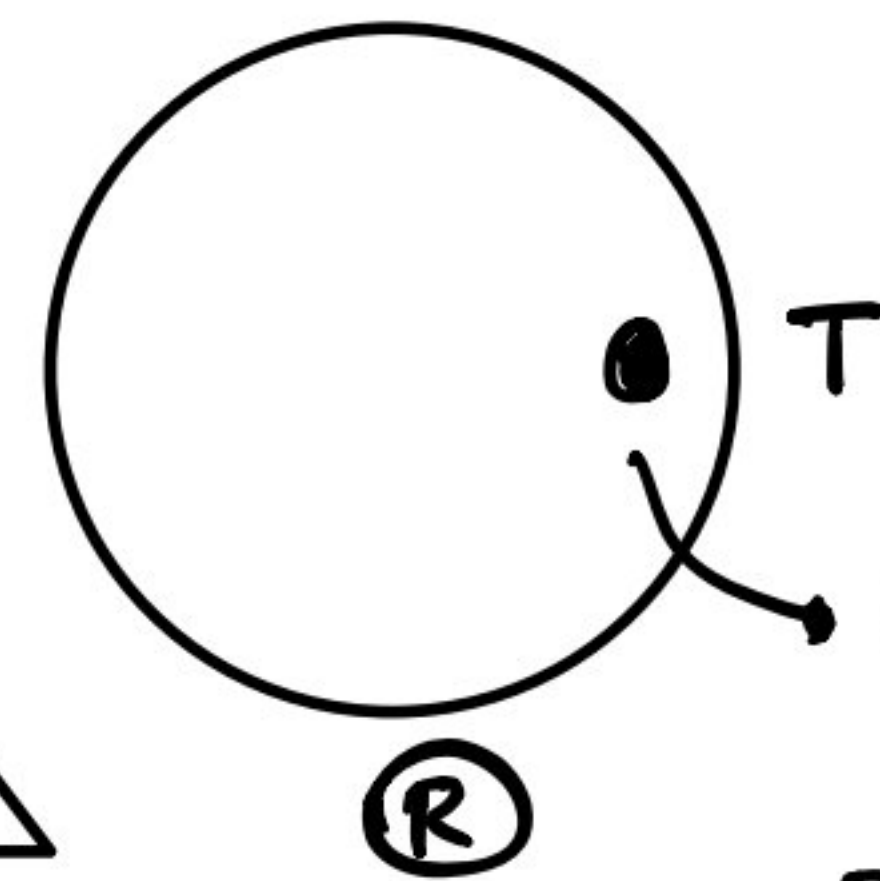
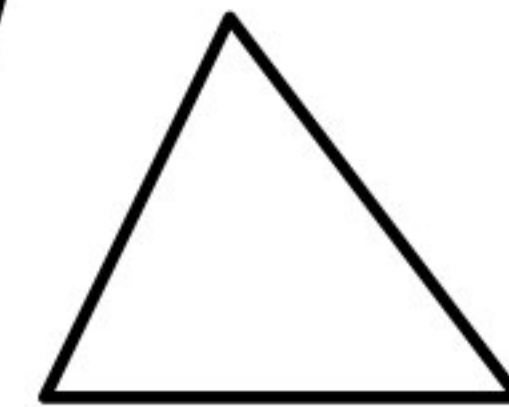
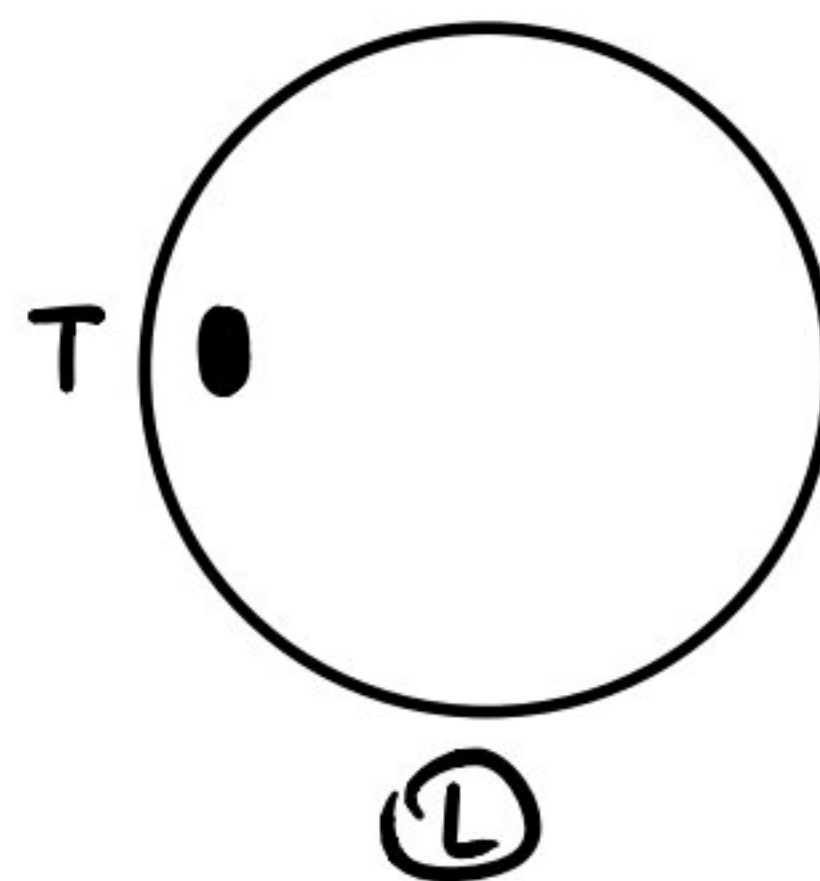
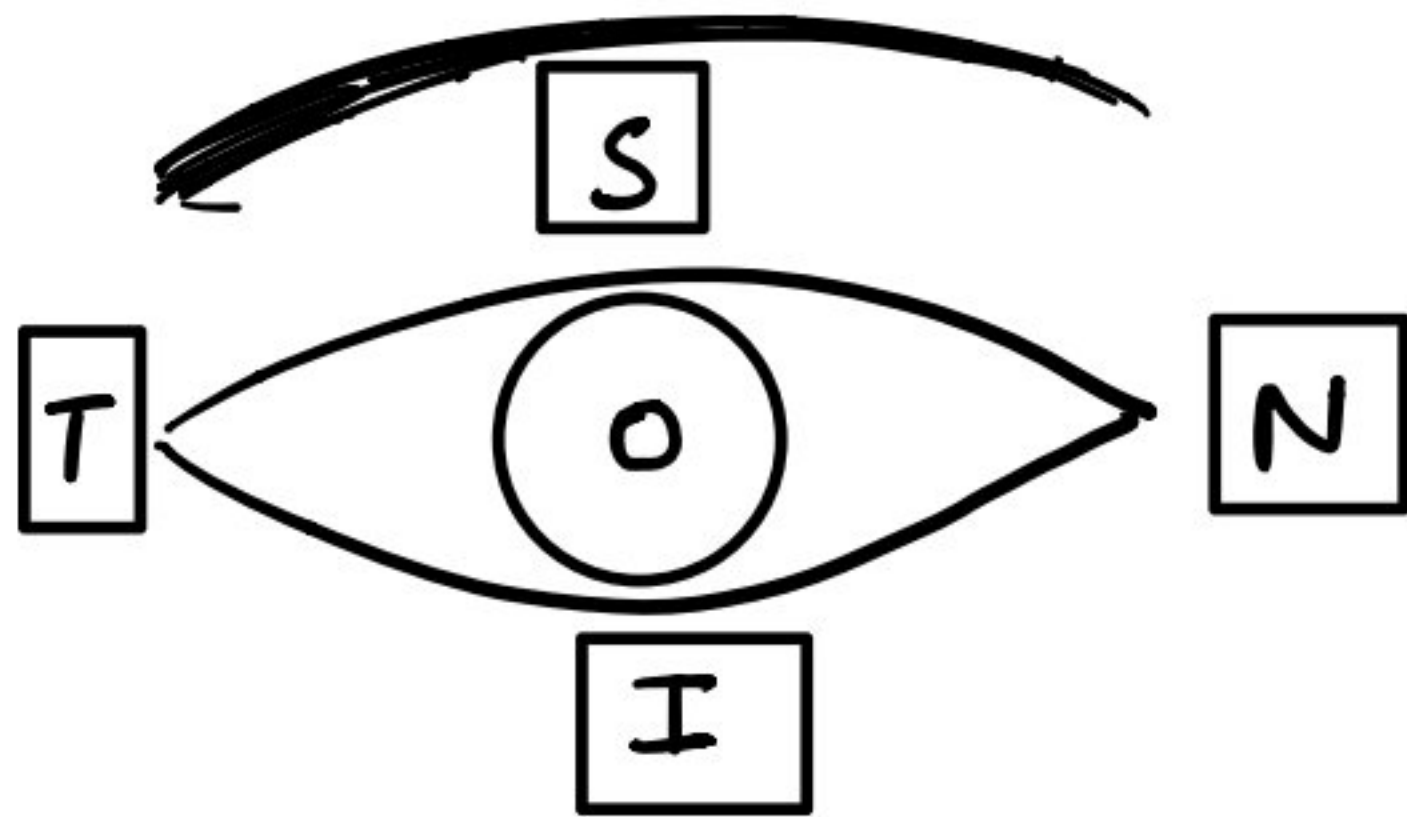
i) Non contact : +90D / +78D

ii) Contact : Goldmann 3 mirror lens → +60D



Visual field:

• maximum visual field @ Temporal



Optic disc
↓
Blind spot
↓
Negative scotoma

Perimetry

Assessment of visual field

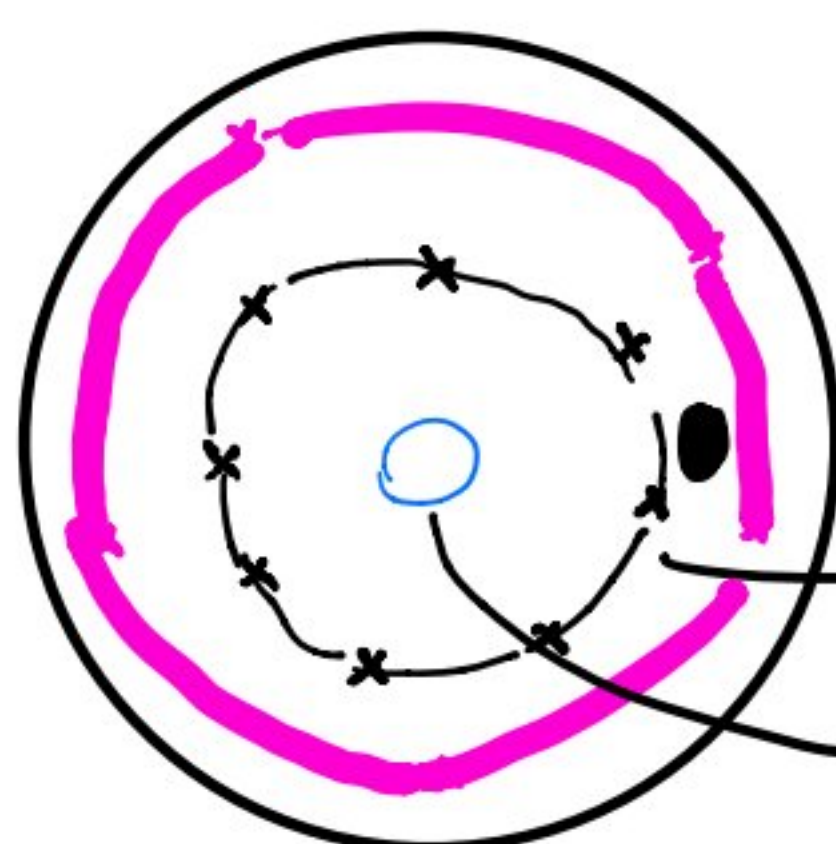
Kinetic Static

Kinetic

• Light target is moved from non seeing to seeing area
• Repeated using different target sizes.

Isopter :

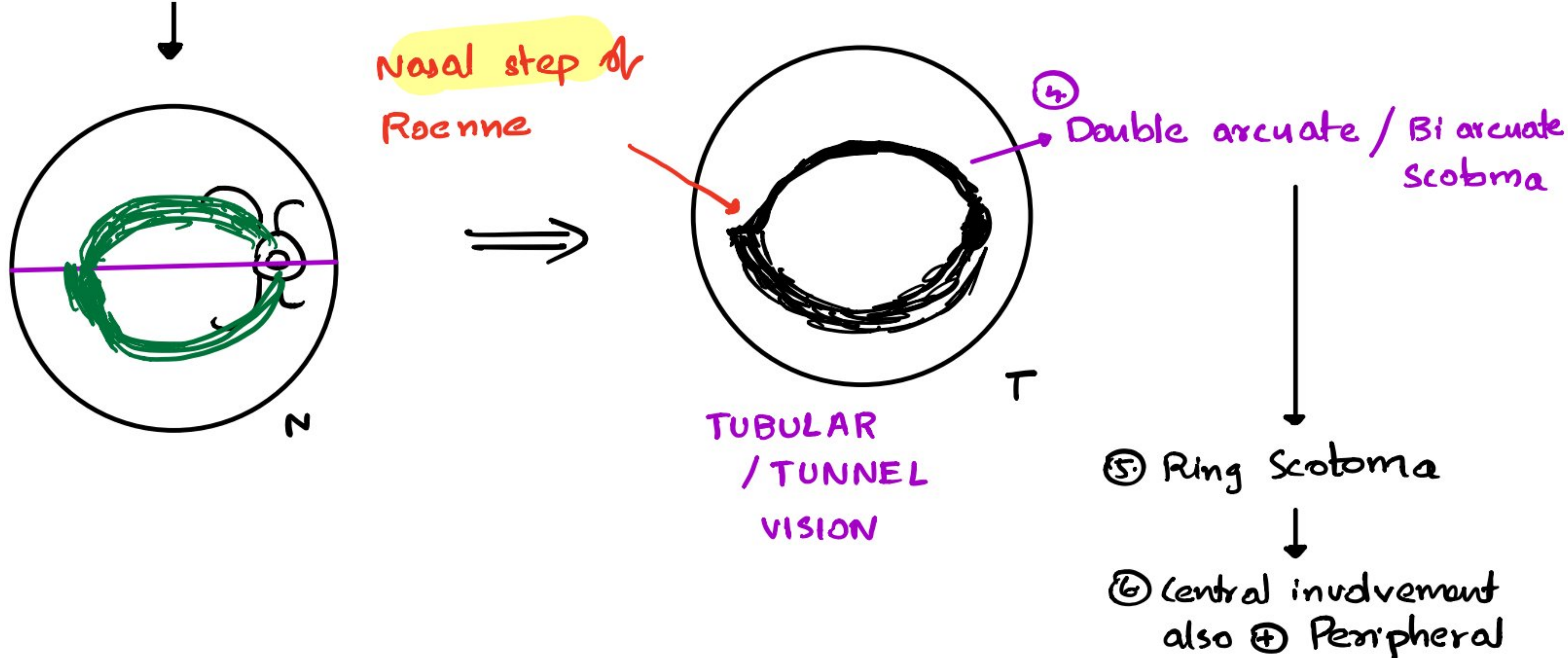
NORMAL



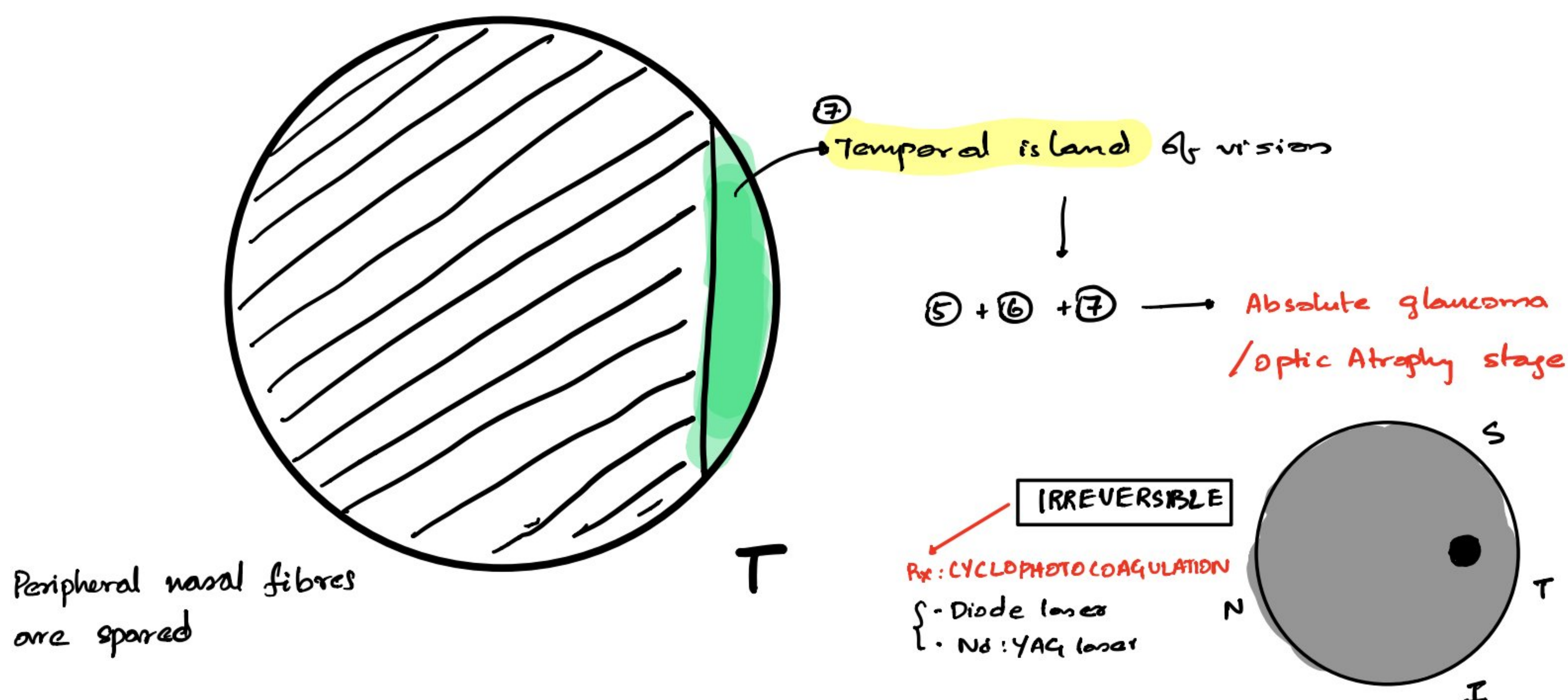
visual field

isopter : the curve joining same target size.

isopeter at central retina.



Advanced / Endstage Glaucoma



KINETIC PERIMETRY

✓
LISTER'S Perimeter
Peripheral field

BIERRUM'S tangent screen
Central field

GOLDMANN'S PERIMETER

• Both peripheral & central field assessment.

Clinical method to assess visual field : Confrontation test & type of kinetic perimetry
↓
{Finger movement} → Bedside assessment

STATIC PERIMETRY :

- Light target donot move
- Target size → same
- Light intensities keep changing @ each testing point.

Types

Humphrey's Automated perimetry

OCTOPUS PERIMETRY {manual}



Humphrey's automated perimetry

→ Goldman size 3 light used

Gold standard perimetry

→ Goldmann perimetry - (manual)

Kinetic
Static

POAG:

ETIOPATHOGENESIS (GLC1 defect)

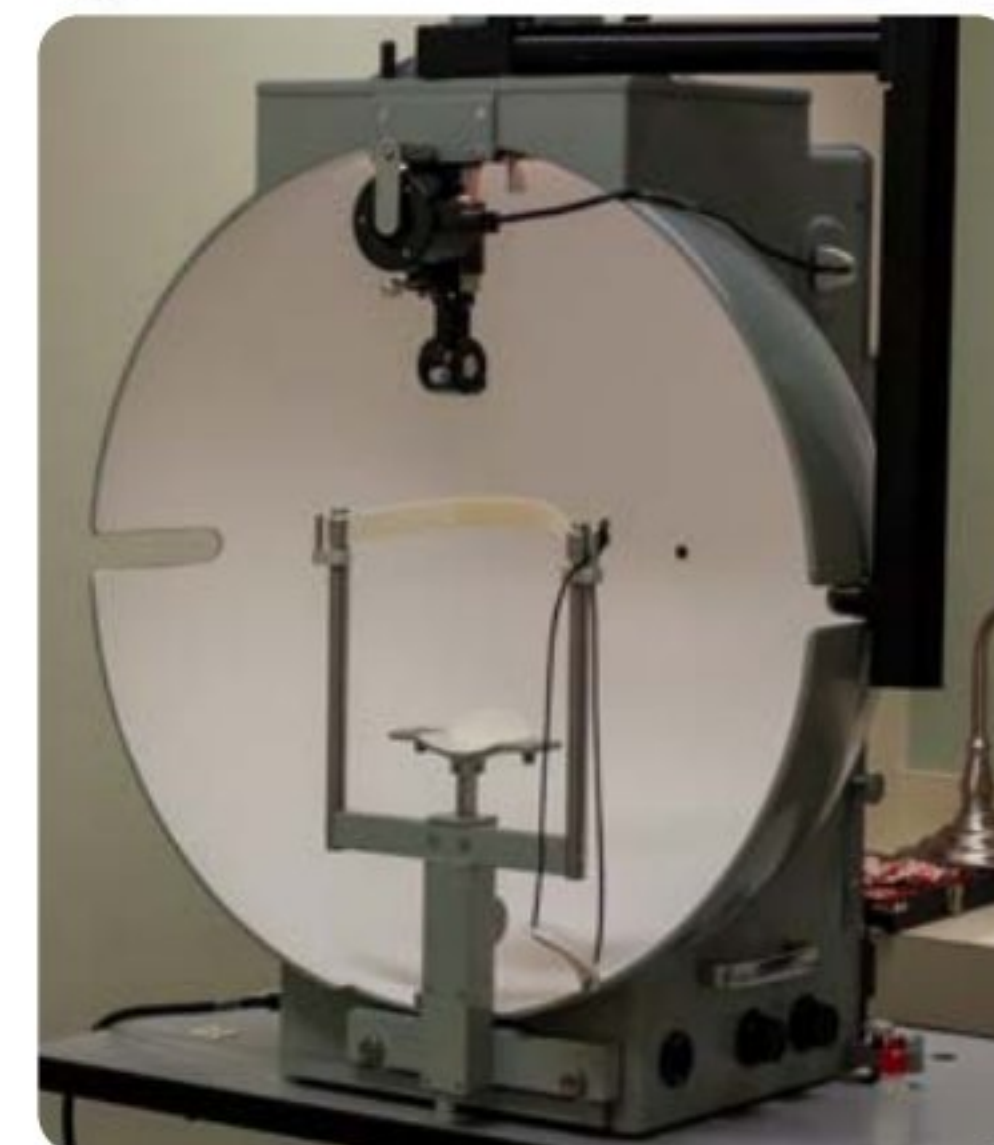
- i) Hereditary (Multifactorial), family history
- ii) Myopes - more common
- iii) Hypertension - more common
- iv) Rise in IOP → due to decreased outflow due to thickening and sclerosis of Trabeculae
- v) Black race
- vi) Age → common in elderly
- vii) Diabetes - more common
- viii) sex male > female

PACG:

Predisposing Risk factors

- i) Hypermetropic eyes with shallow anterior chamber
- ii) Iris - Lens diaphragm placed anteriorly
- iii) Narrow angle of anterior chamber
- iv) Age - more common after 5th decade
- v) Sex - females are more prone
- vi) Positive family history.

Lister's perimetry



Goldmann's perimetry

Provocative Test

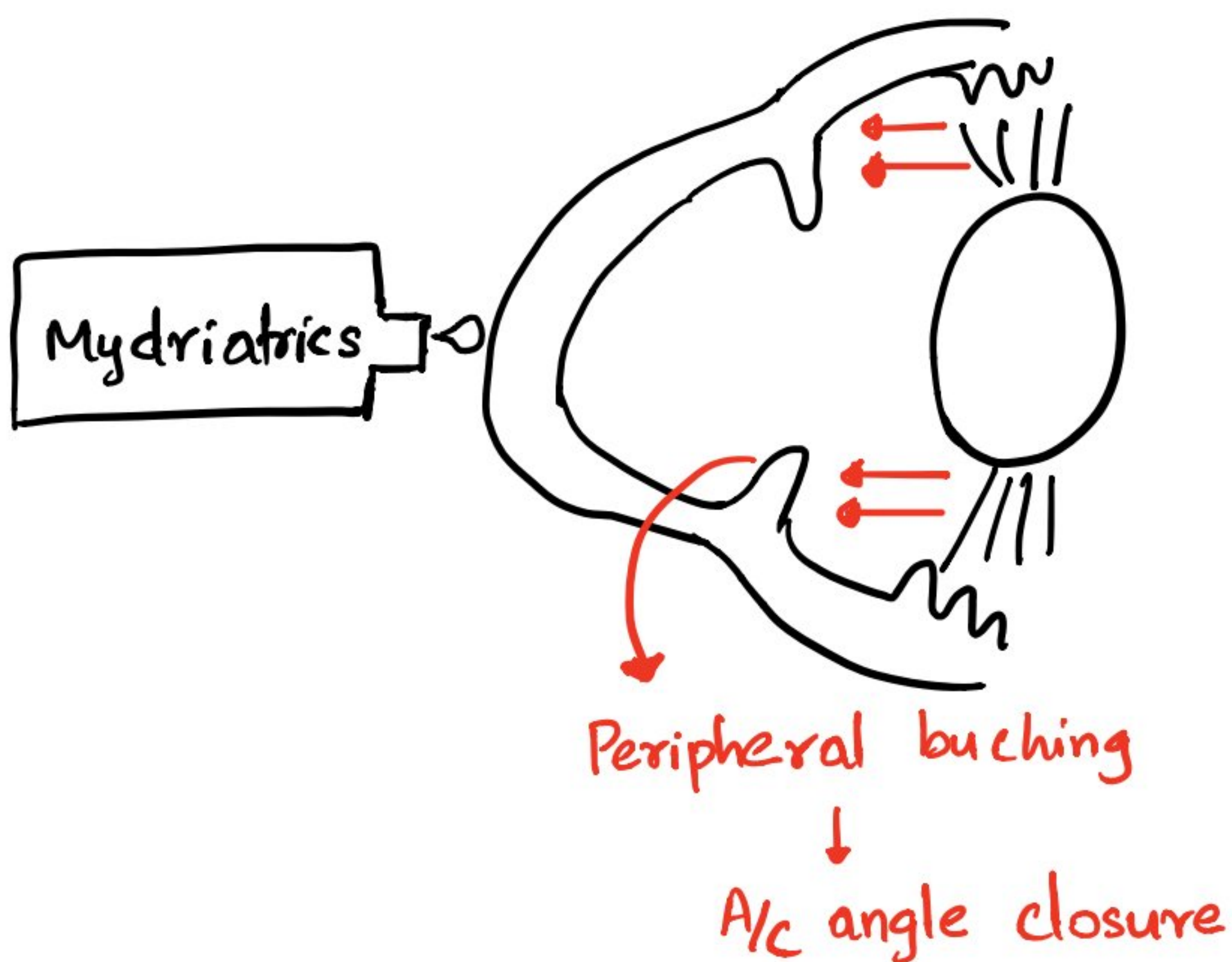
for POAG

is Water drinking test

Pre-test IOP $\xrightarrow{1\text{hr}}$ Post test IOP

IOP rise $> 8\text{mmHg}$ $\rightarrow$ High chance

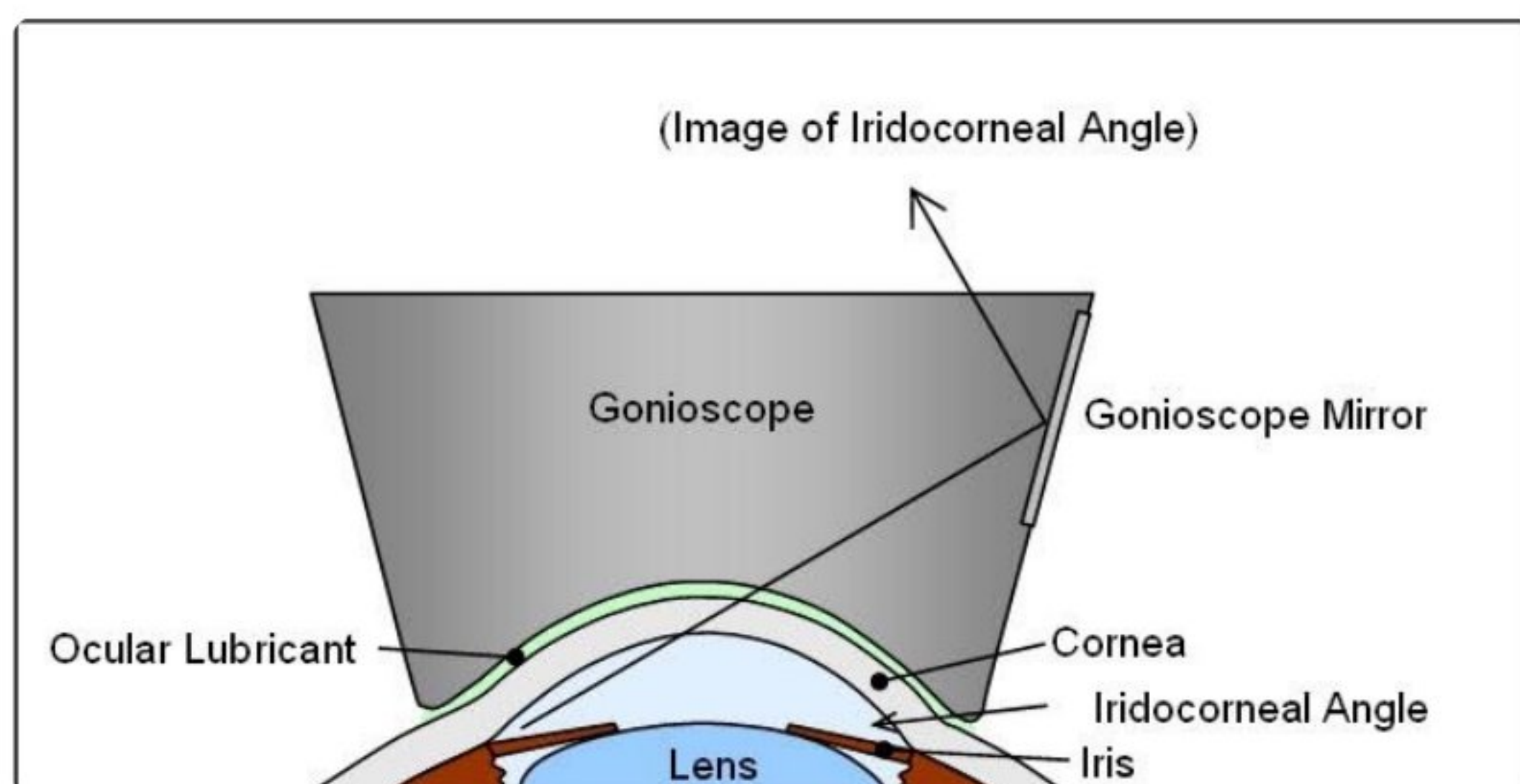
for PACG



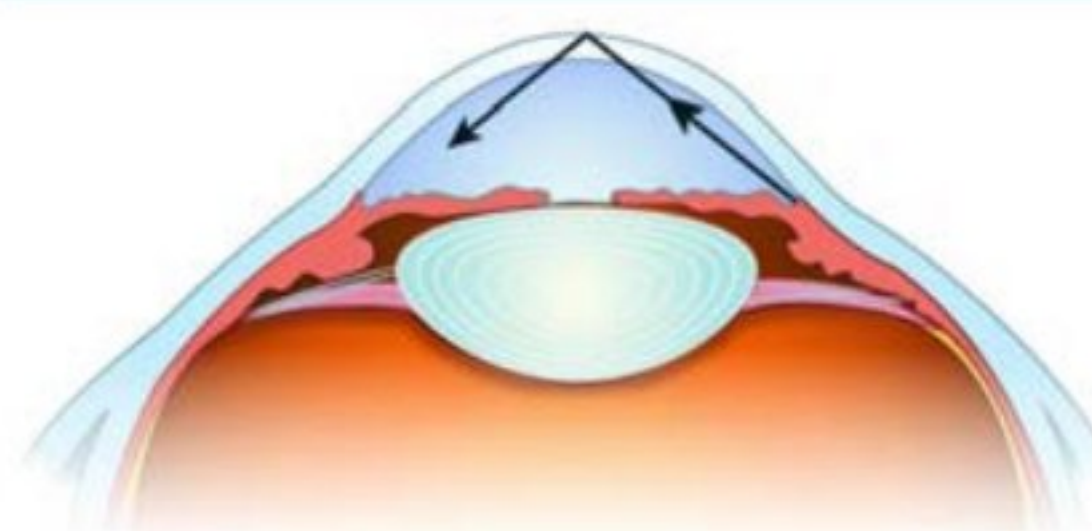
- 1) Dark room test (for 1hr) $\Delta\text{IOP} > 8\text{mmHg}$
- 2) Prone test $\{ \text{micro movt of Iris forward} \}$
- 3) Prone - Dark room test (1hr) $\rightarrow$ pt. shud not sleep $\because$ In sleep miosis happens
- 4) Mydriatic test
- 5) Mydriatic - Miotic test $\rightarrow$ mid dilation achieved most reliable

Primary Glaucoma $\rightarrow$ POAG (MC) worldwide
 $\rightarrow$ PACG (MC) in Asians

Gonioscopy $\rightarrow$ Visualisation of the angle



Critical Angle
Cornea Air Interface ~ 46 degree Light rays from Angle exceeds Critical angle so rays reflected back into AC (Total Internal Reflection), preventing direct visualisation of angle



• Normal slit lamp angle cannot be visualised d/t

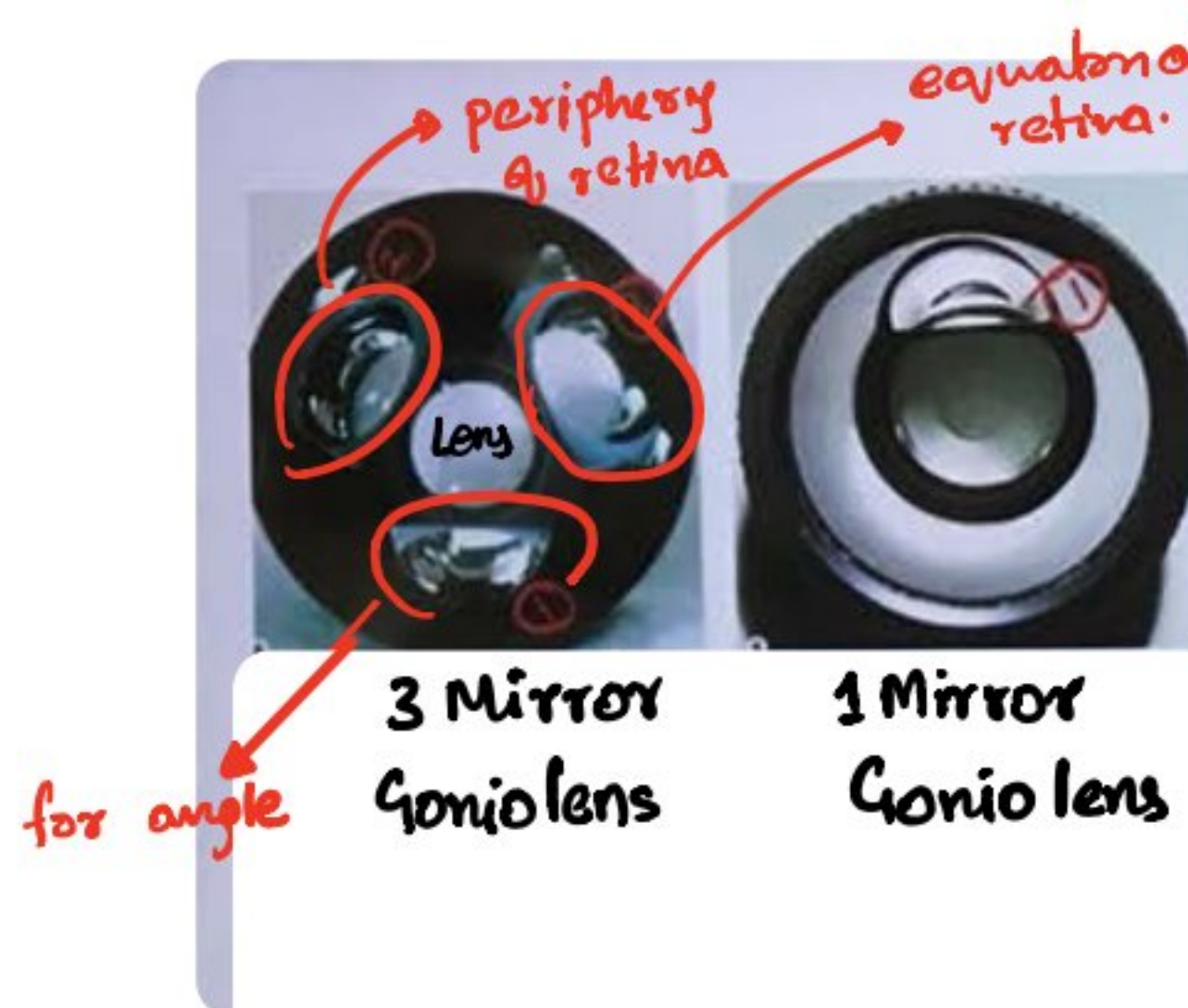
Total Internal Reflection

$\downarrow$
Critical Angle = 46°



4- mirror gonioscope

UNDILATED PUPIL



GOLDMAN GONIOMIRROR

View of Gonioscopy

Angle structure

⇒ (P) 9

Can

Iris Root

Ciliary Body Band

See

Scleral spur

Till

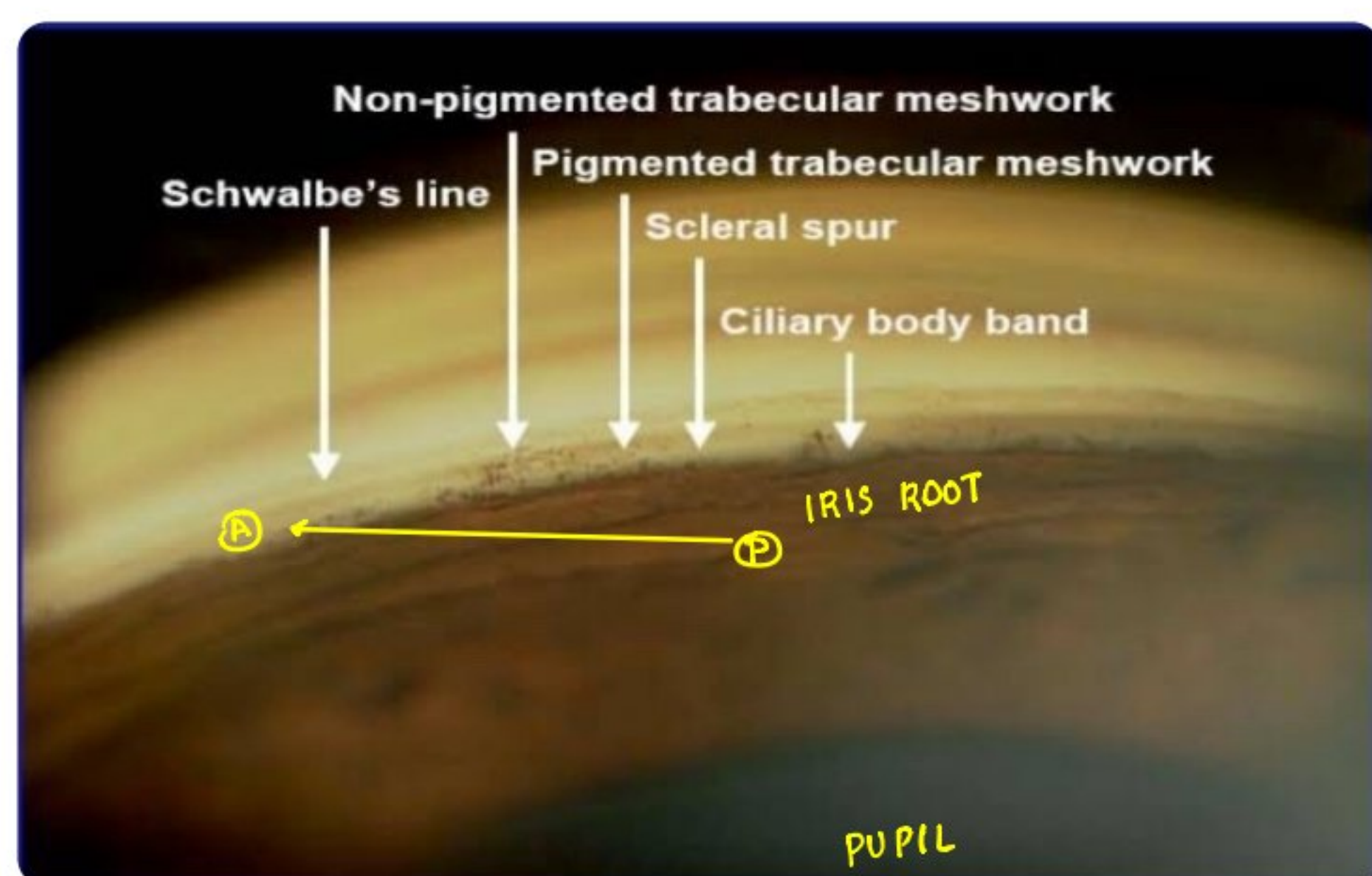
Trabecular meshwork ✓

(A)

Shwalbe

shwalbe's line ✓

at least up to hear should be viewed to call "the angle is open"



Q: All of the following are true about the instrument shown in the image given below except that _____



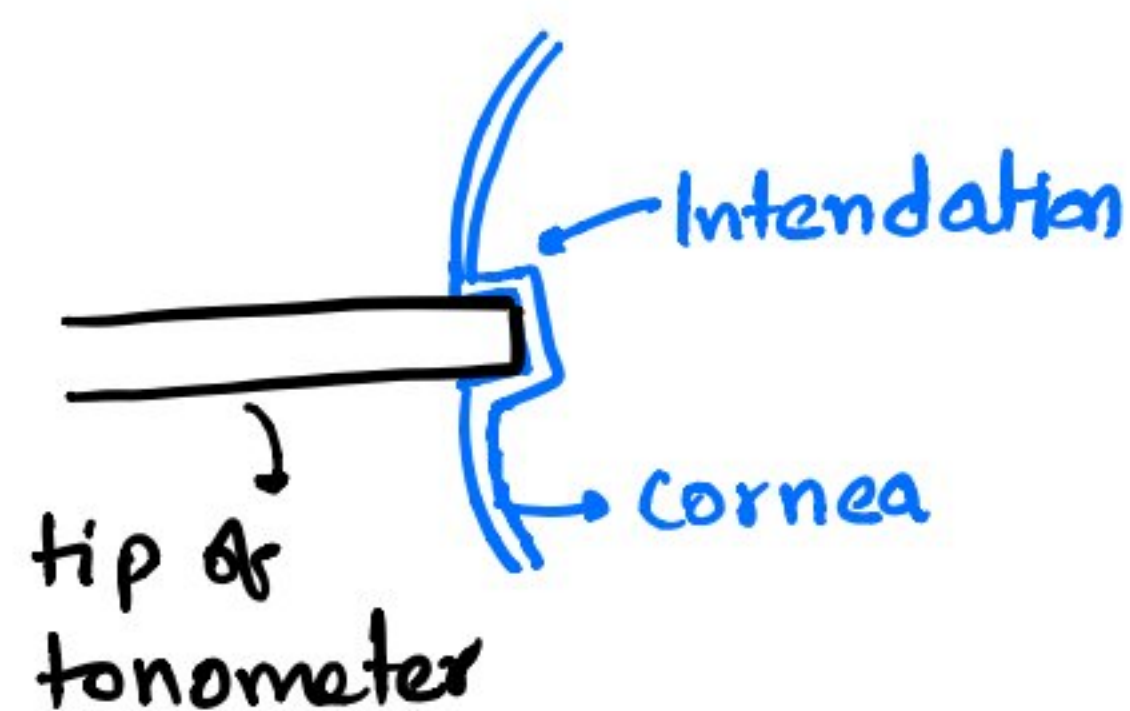
- a) It is used for the localization of foreign body.
- ☒ b) It is used for the measurement of the angle of the posterior chamber.
- c) It is used for the demonstration of the extent of the peripheral anterior synechiae
- d) It provides a mirror image of the opp. angle.

TONOMETRY : measurement of IOP

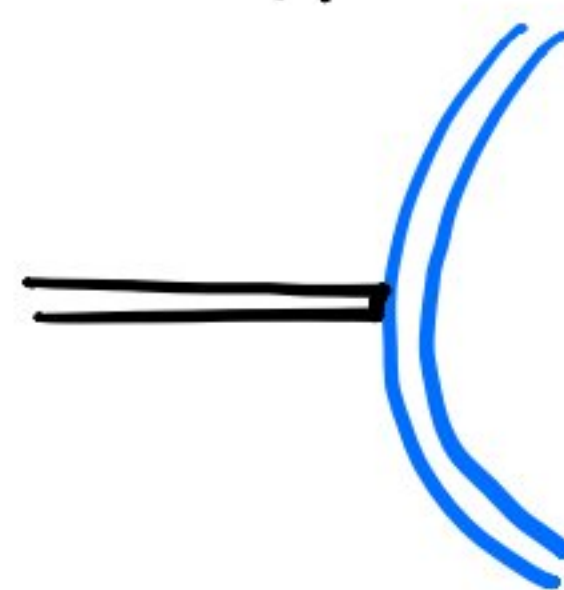
- IOP generated d/t resistance of (N) aqueous outflow
- Resistance max @ Juxta canalicular position of TM.
- (N) IOP = 10-21 ⇒ 16mmHg
- IOP is diurnal variation: upto 8mmHg
- max. IOP: early morning (4:00am)

Types of Tonometer

Indentation



Applanation



Type of INDENTATION TONOMETER

Eg: ~~of~~ SCHIOTZ TONOMETER

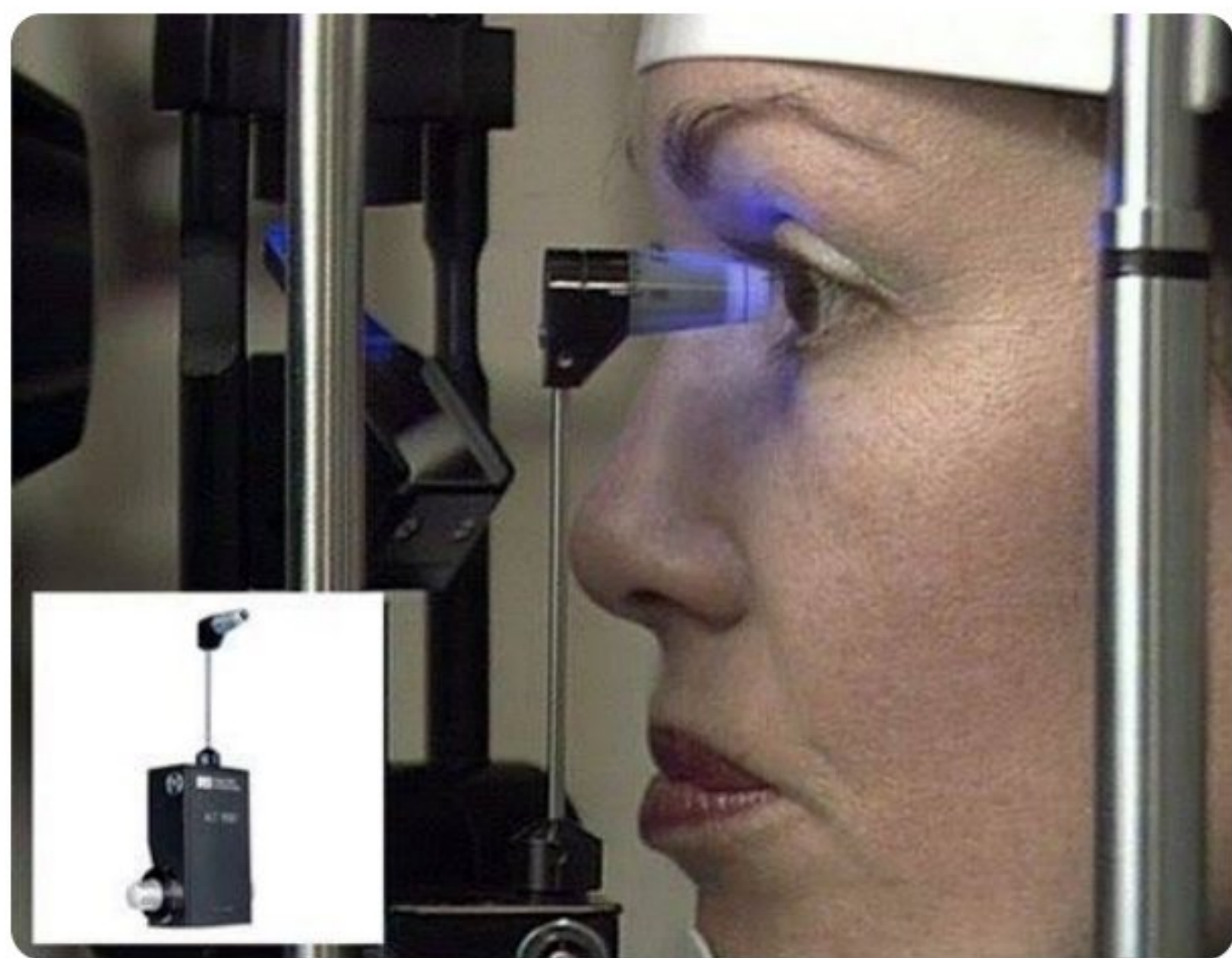


- Supine position
- Variable force/weight
5.5g / 7.5g / 10g
- Scale reading wght → FREIDANWALDS nomogram

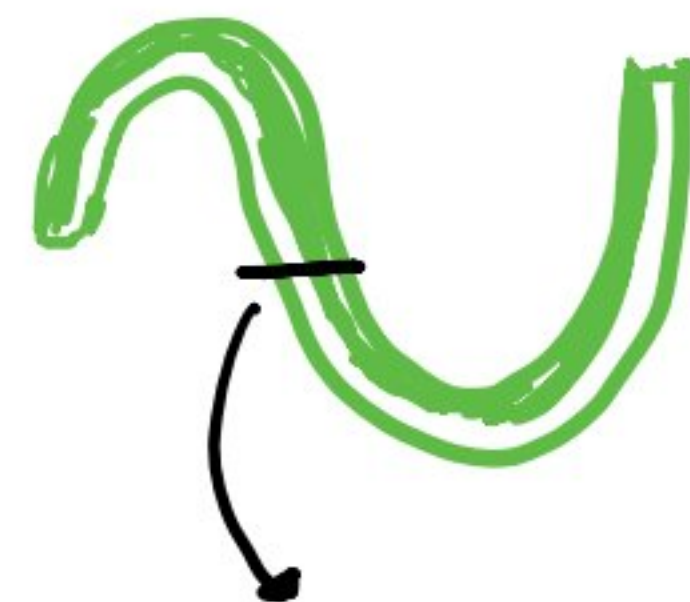
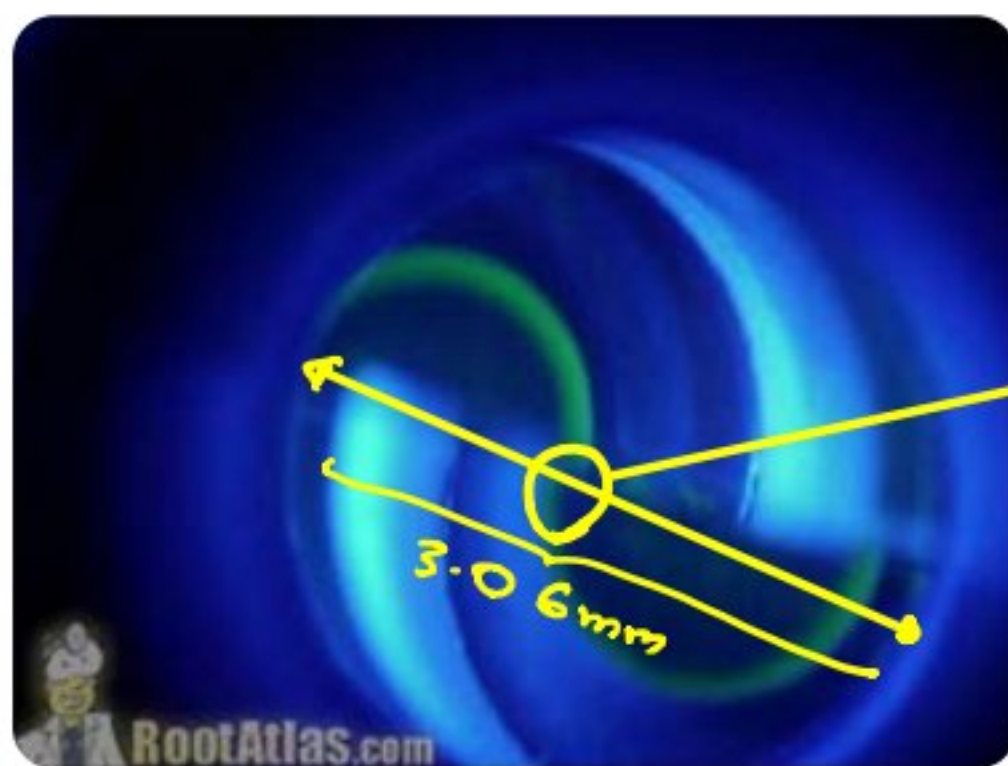
Disadvantage:

- 1) (MCC) of spread of Epidemic conjunctivitis
- 2) IOP reading affected by scleral thickening
- 3) Myopia → falsely low IOP.

TYPES OF APPLANATION TONOMETER:



- Based on Imbert - Fick Law
- Attached on slit lamp
- Consist of two prism
- Topical Fluorescein dye → Cobalt blue light



when inner edge of two semicircles meet → correct IOP

- Gold standard tonometer → Goldmann Applanation
- Affected by central corneal thickness

10 μ difference = 1mm Hg IOP dibt.

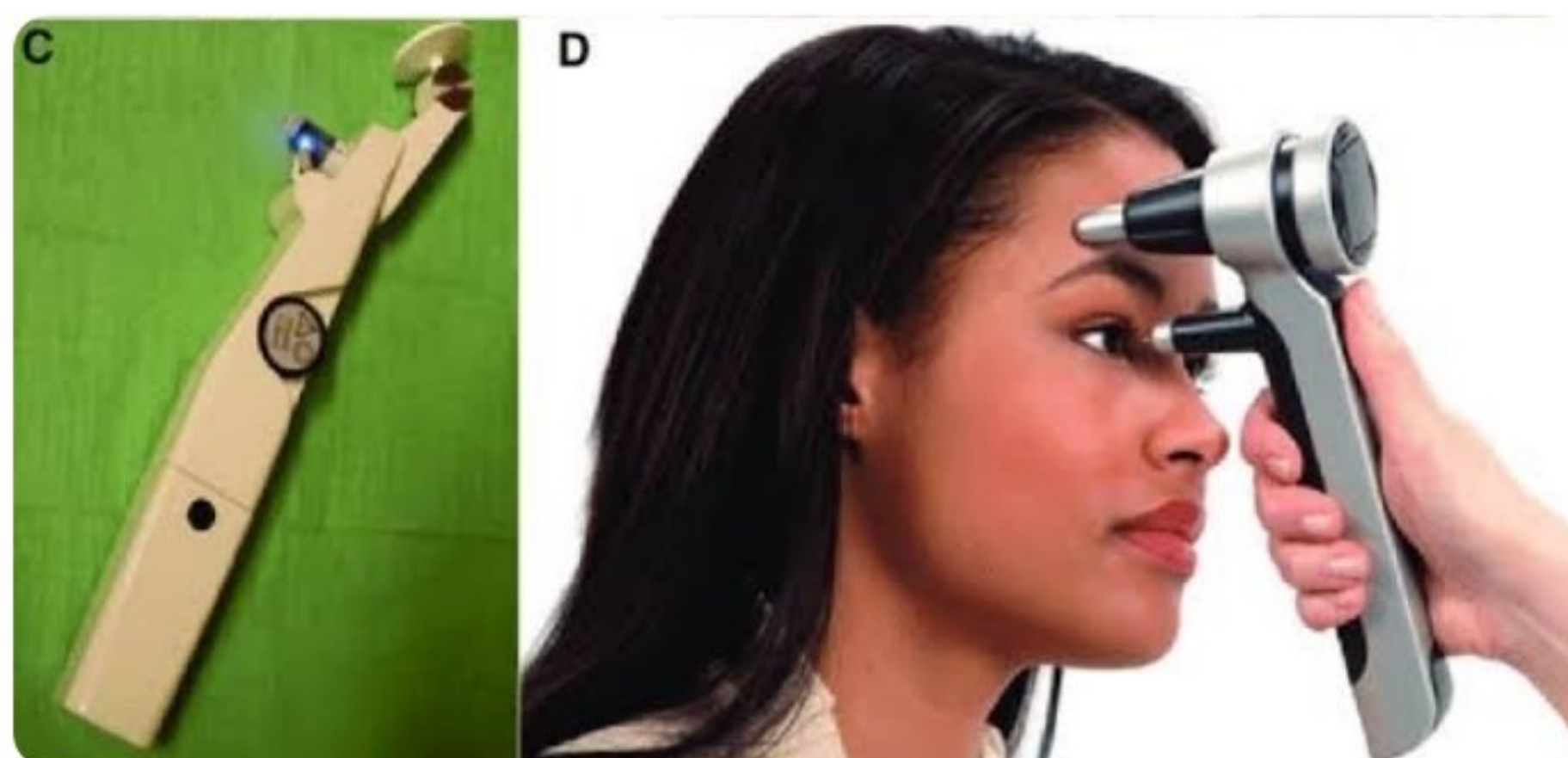
- IOP = 26 mmHg
- Pachymetry = 580 μ
- Reference value of CCT (central corneal thickness) : 520 μ
- [580 μ - 520 μ] 60 μ = 6 mm Hg (dibt) → cIOP = 20 mmHg

- Thick cornea = False High IOP
- Thin cornea = False low IOP
- Flatten contact fixed area $\Rightarrow$ 3.06mm of cornea

$$\text{Area} = \pi r^2 = 7.35 \text{ mm}^2$$

- Tonometer tip cleaned by using isopropyl alcohol after each use and sodium hypochlorite once in a while.

2) PERKIN'S APPLANATION TONOMETER



- Same as goldmann
- Adv: Handy / portable
- Used in IOP assess for congenital Glaucoma

3) TONOPEN



- out patient
- user friendly
- digital

Q: measure IOP in a scarred cornea

- ## 4) MACKAY MARG tonometer :
- can be used in scarred cornea
 - Variable area & fixed weight

NON- CONTACT APPLANATION TONOMETER

- ### 1) Non contact air puff tonometer $\rightarrow$ (Pulsar)
- No need of topical anaesthesia
 - Screening of glaucoma $\rightarrow$ Best
 - Post op patient after glaucoma surgery

NEWER TONOMETER

1) Rebound tonometer

"Icare"

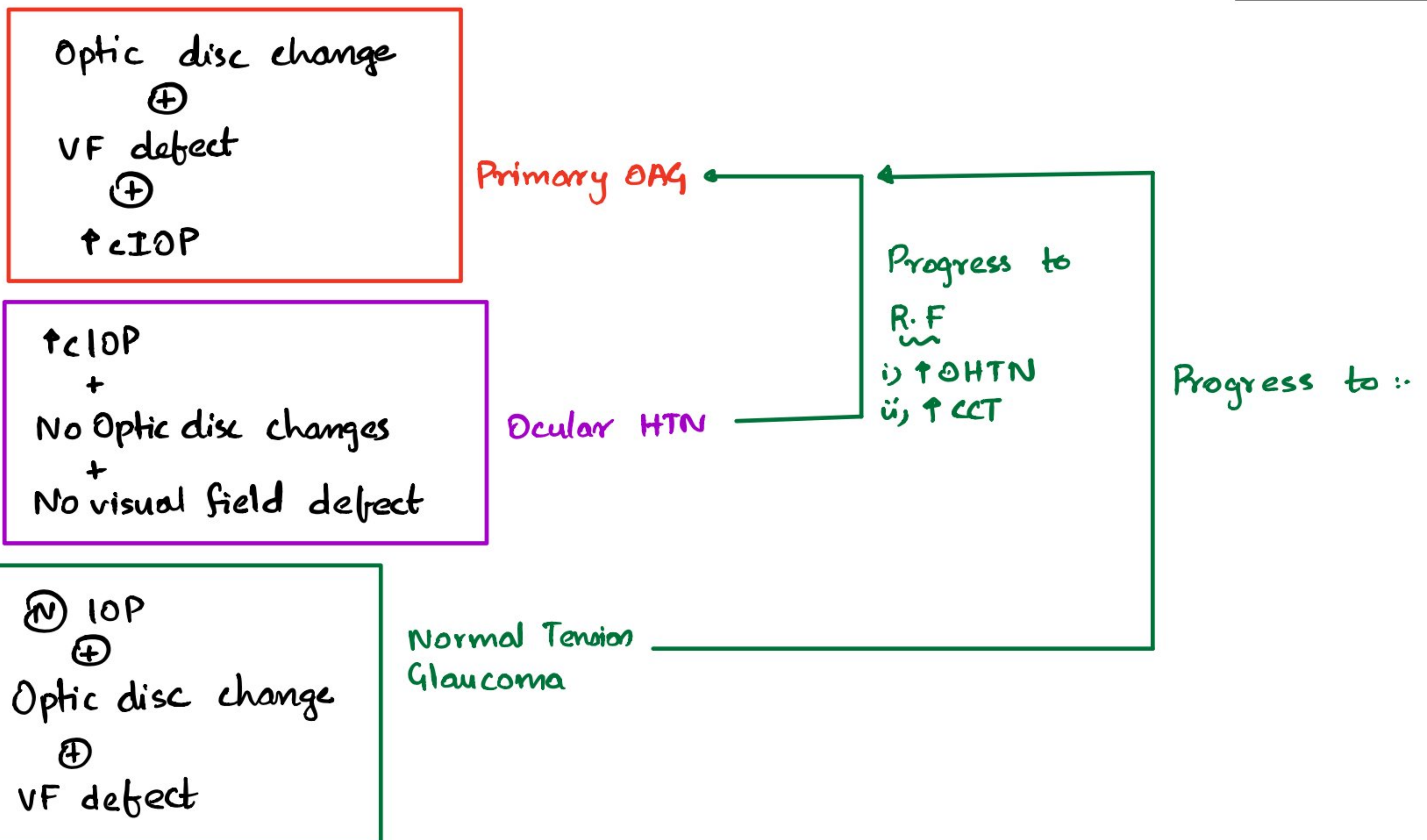
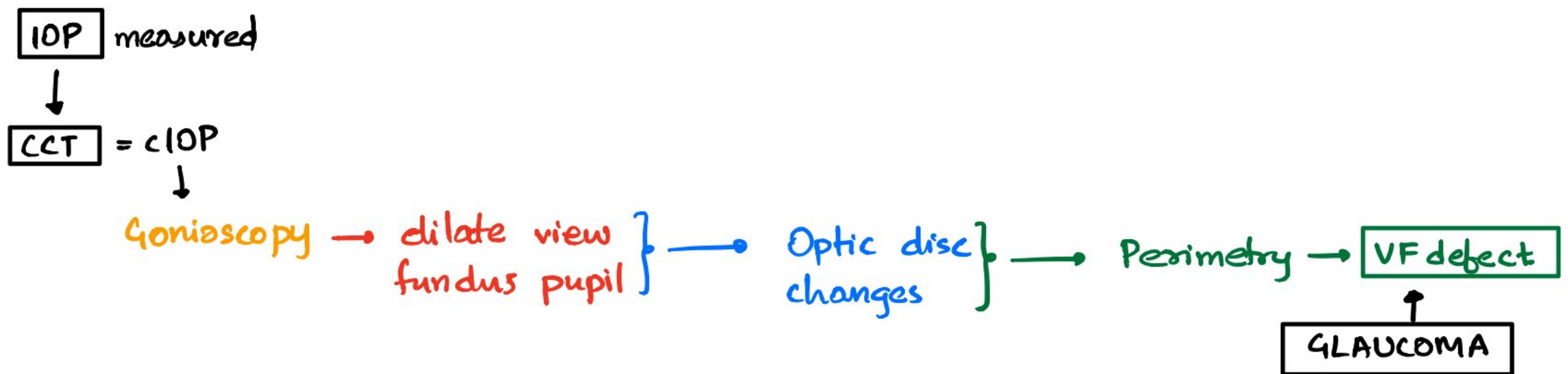


- solenoid
- contact type (brief)
- No need for topical anaesthesia
- self use @ home.

2) Transpalpebral Tonometer : used over closed eyelid **"DIATON"**

- 3) Dynamic Contour Tonometer (DCT) →
- Pascal
 - not affected by CCT
 - Principle: "Contour matching"
 - Gives corneal hysteresis: Biomechanical properties of cornea like elasticity
↳ used in keratoconus.
- 4) Ocular Response analyser: Not influenced by CCT
• Ocular blood flow value by "OCULAR PULSE AMPLITUDE"

Algorithm for Glaucoma test



Tonography: measurement of FACILITY of Aqueous outflow & plotted on graph?

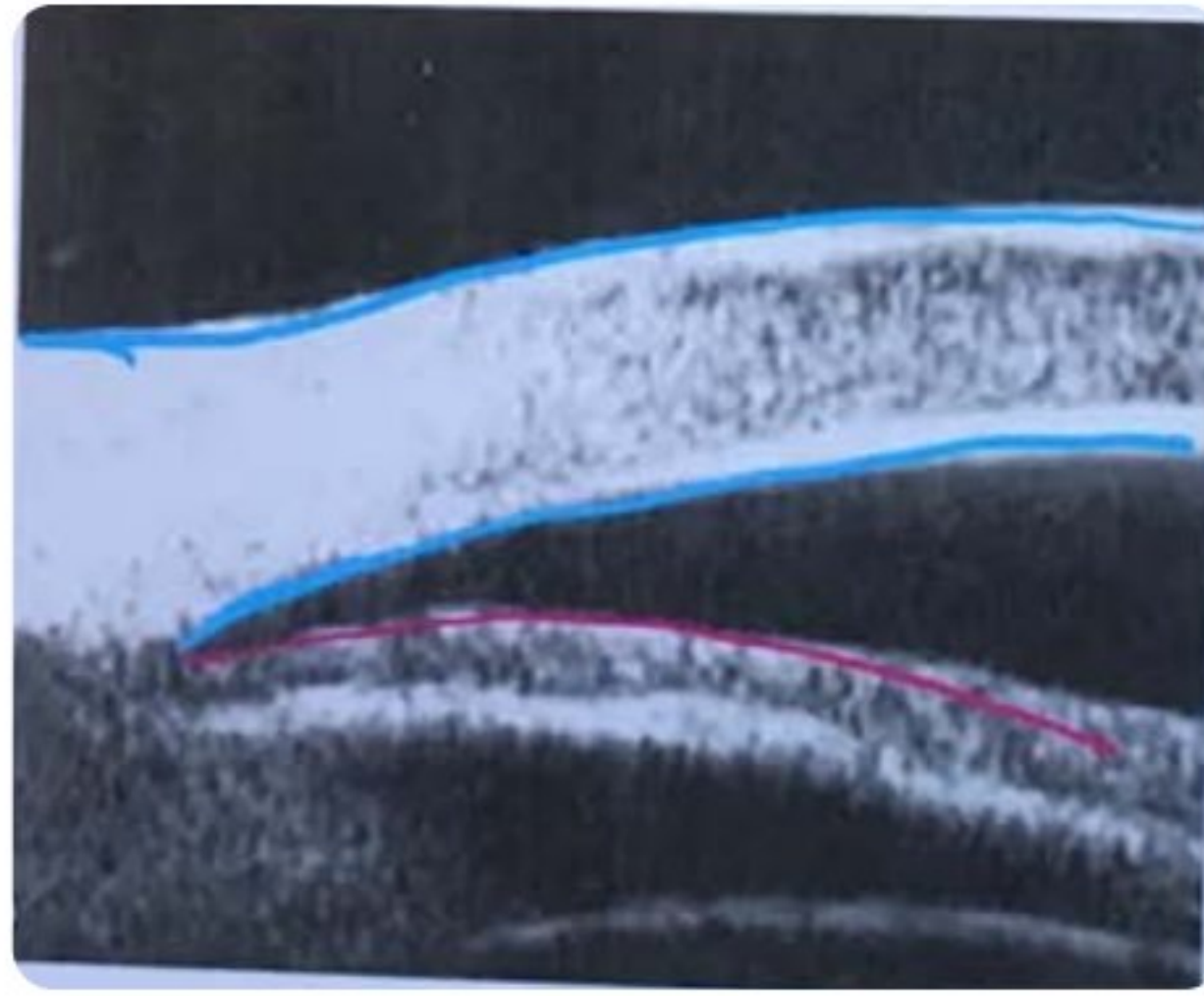
Primary Angle Closure glaucoma:

Indirect assessment of angle →

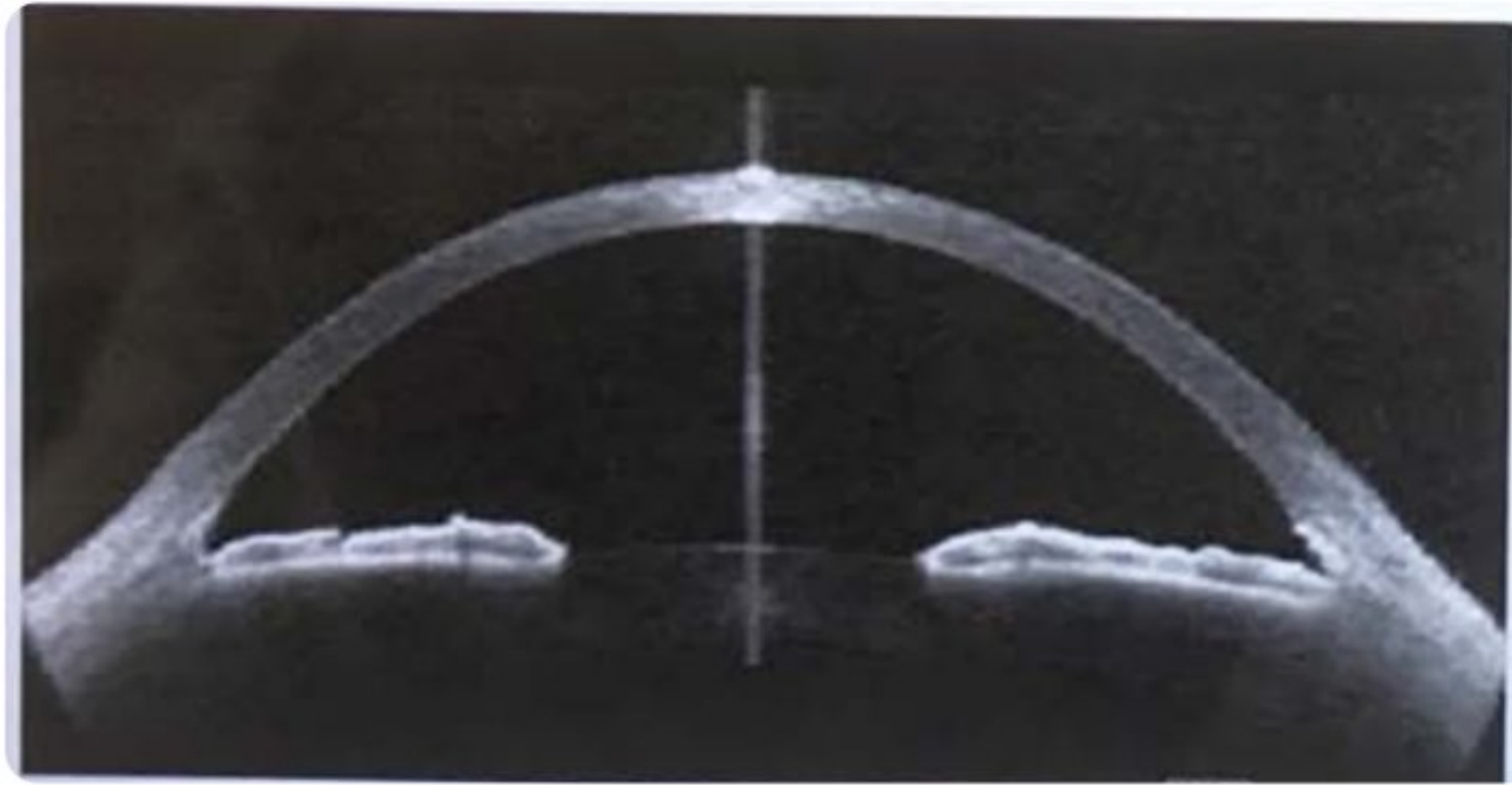
- i) Ant. chamber depth
- @ level of limbus
- compare corneal thickness & ACD & according to beam thickness?

} VAN HERICKS METHOD

ii) USG Biomicroscopy $\Rightarrow$ 35-50Hz {High}
 (N) USG = 8-10 Hz

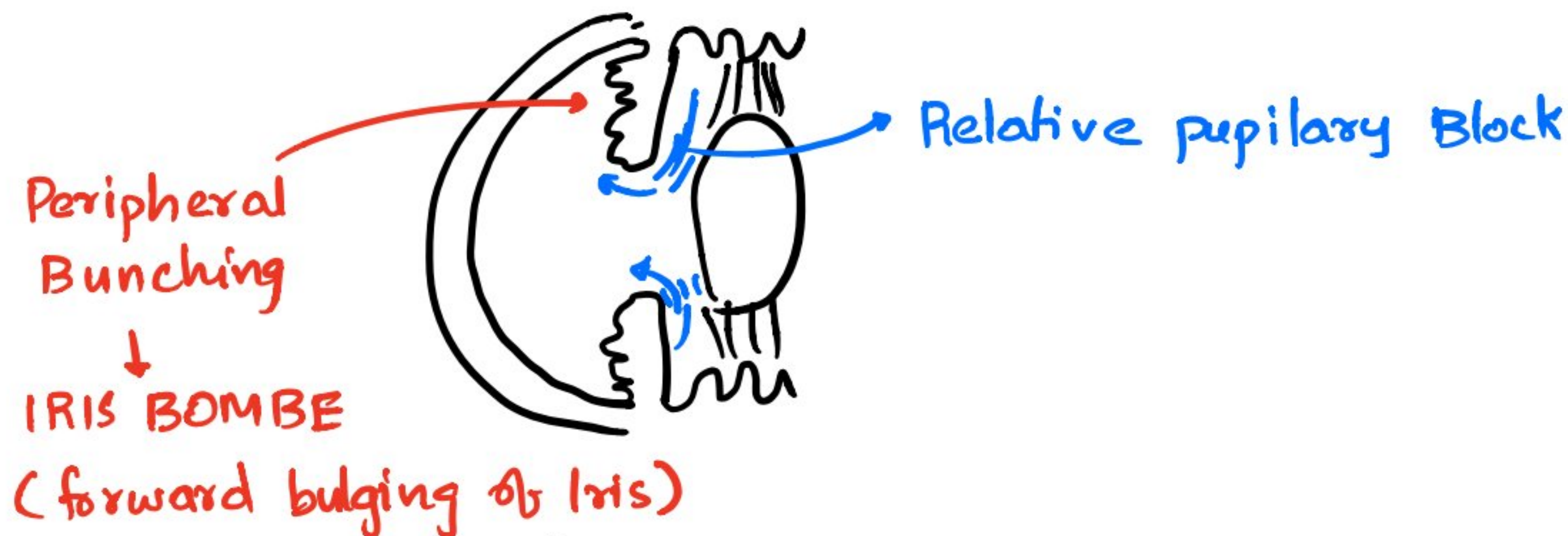
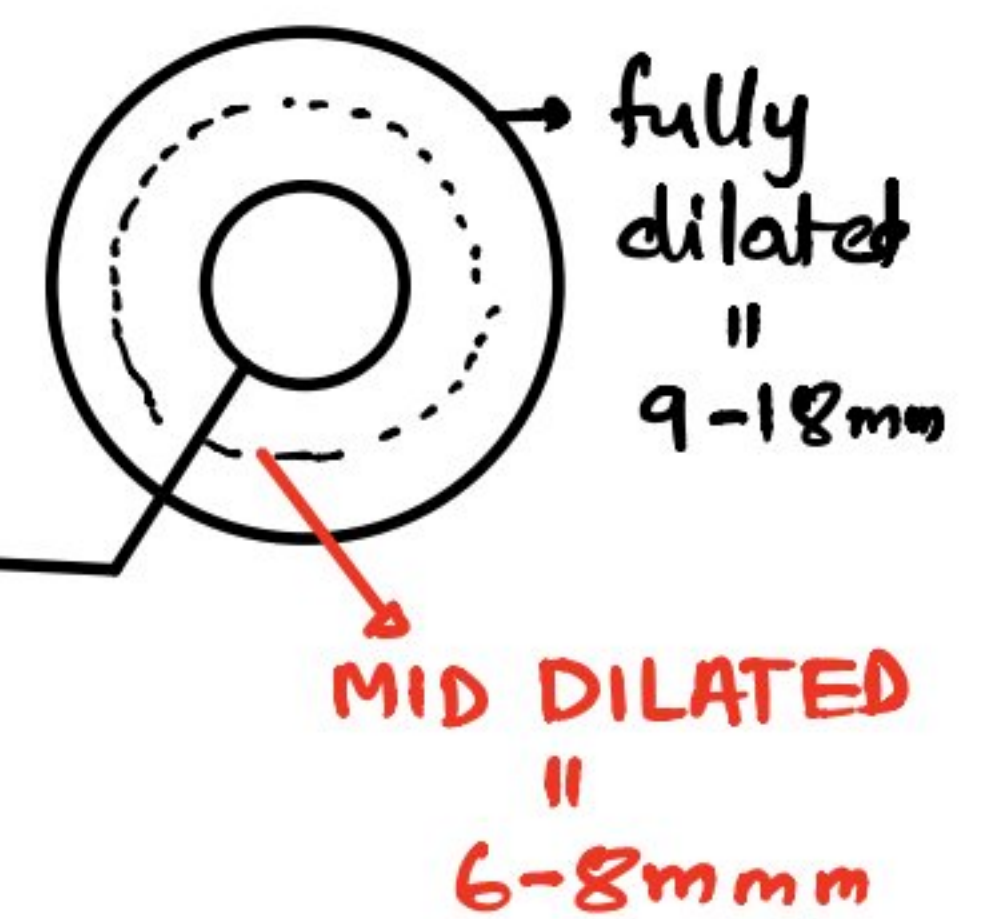


iii) AS-OCT

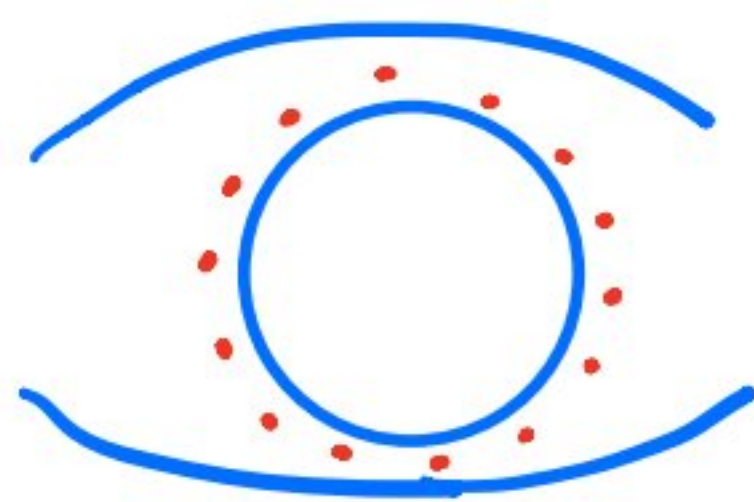


A/c attack of angle closure / A/c congestive glaucoma

• D/t mid dilated pupil $\xrightarrow{\text{S/I}}$ i, Reading in dim light
 ii, Cinema theater



= Rapid $\uparrow\uparrow\uparrow$ IOP
 40-60mmHg



- AC shallow
 - Angle closed
 - Tonometry = IOP $\uparrow\uparrow$
 - Pain
 - Red Eye
 - Sphincter Muscle Ischemia
- ↳ Blood congestion
 ↳ stretching optic N
 ↳ Vertically oval pupil

Folds in the iris $\rightarrow$ folds back on the lens $\rightarrow$ $\uparrow$ IOP $\rightarrow$ forces the post. chamber

Pushes the angle $\leftarrow$

↓
IOP rises in 30mins (x 4 times)

15 mm Hg → 30 mm Hg

loss of vision, vomiting d/t pain ⇒ A/c angle closure
glaucoma / acute
congestive glaucoma

VOGT'S TRIAD

Pupil findings ⇒
1) Mid dilated pupil
2) Vertically oval
3) Non reactive to light.

* Endothelial pump failure → water accumulates to stroma } ↓ vision
⊕ Colored halo ^{4T-VI}

• Cornea → Hazy & d/t failure of endothelium pumps

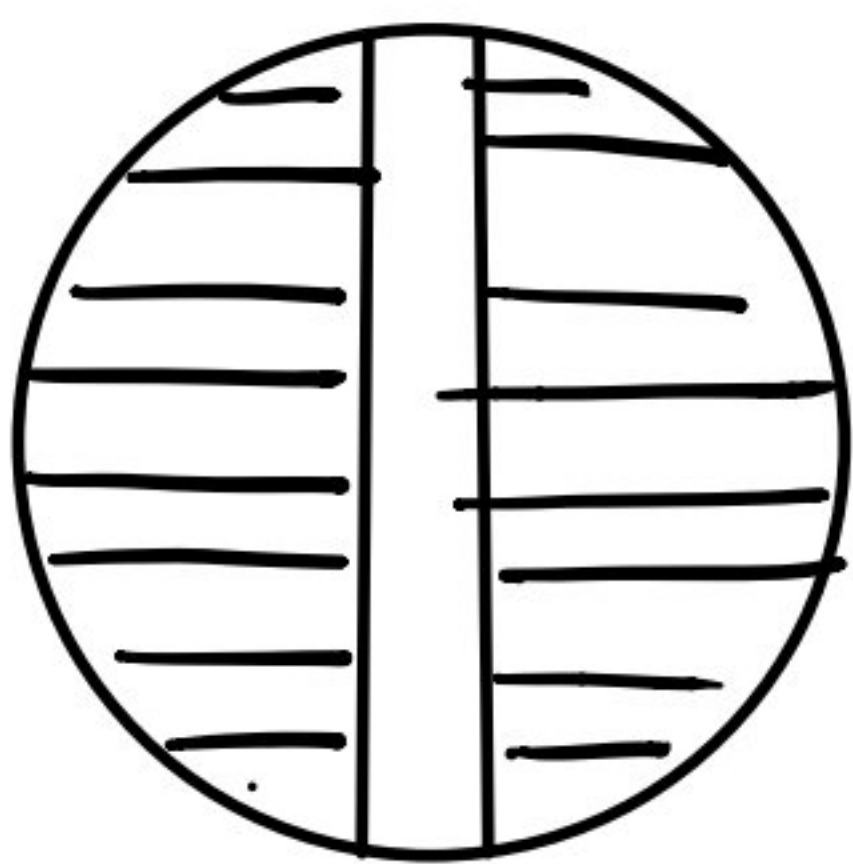
∴ Presentation in A/c congestive → Redness, painful decreased vision
⊕ coloured halo

Symptoms ∴

- Pain (stretching of optic nerve)
- Coloured haloes (corneal edema)
- Sudden drop in vision.

Distinguish coloured haloes of Glaucoma & cataract

FINCHAM'S TEST



Cataract
• Split up &
write again

ACG
• Do not split

Permanent change after A/c attack ∴

1) Ant. sub capsular fibre necrosis → ASC opacity k/a GLAUCOM FLECKEN
& d/t stagnant aqueous → Toxic product
accumulates in front of lens

2) Ischemic sphincter → sphincter atrophy a.k.a
IRIS ATROPHY

3) Melanin from iris & d/t repeated constrict ↔ dilation
deposits on corneal endothelium → k/a KRAUKENBERG SPINDLE



Glaucomflecken

2nd Vogt's Triad

- i) Glaucomflecken
- ii) IRIS ATROPHY
- iii) Krukenberg Spindle

Choreoretic pupil in ACG

2.



Management of A/c congestive Glaucoma

1st step → ↓↓↓ IOP

2. i) IV mannitol

or

ii) Oral Acetazolamide

2nd step ⇒ MIOTICS

DOC for ACG = Topical PILOCARPINE

TOL : Peripheral iridotomy ⇒ Done for affected eye & prophylactic other eye
for ACG

↳ Laser : Nd : Yag Laser
Photodisruption

↳ Done at superior 10-2'0 clock position of Iris

DOC for PACG: Add PG analogues

Open Angle Glaucoma ⇒ Silent Thief of sight

- Coloured races : Blacks / Africans
- Myopia
- Elderly (60-70s)
- Thin Cornea - faulty IOP
- Trabecular meshwork fibrosis

20-30 yrs → slow rise in IOP

Symptoms :

- Painless
- Cornea is always clear . No corneal edema
- No coloured haloes
- No loss of vision

Visual field is collapsing → damage to optic nerve



TUNNEL VISION / TUBULAR

Only symptom (No other symptom)



- frequent change of presbyopic glasses → OAG
- Frequent change of distance glasses → Cataract (old patient)
- Frequent change of glasses in young → Keratoconus
(N = 1 time a year) Pathological Myopia

ACG	OAG
• Female • 50yrs • Hypermetropia • Sudden, painful • Coloured Haloes • Pupillary Block	• no gender • Older • Myopia • Slow, painless • No symptoms • Trabecular fibrosis

SOME BASIS ON THE DRUG ACTION IN GLAUCOMA

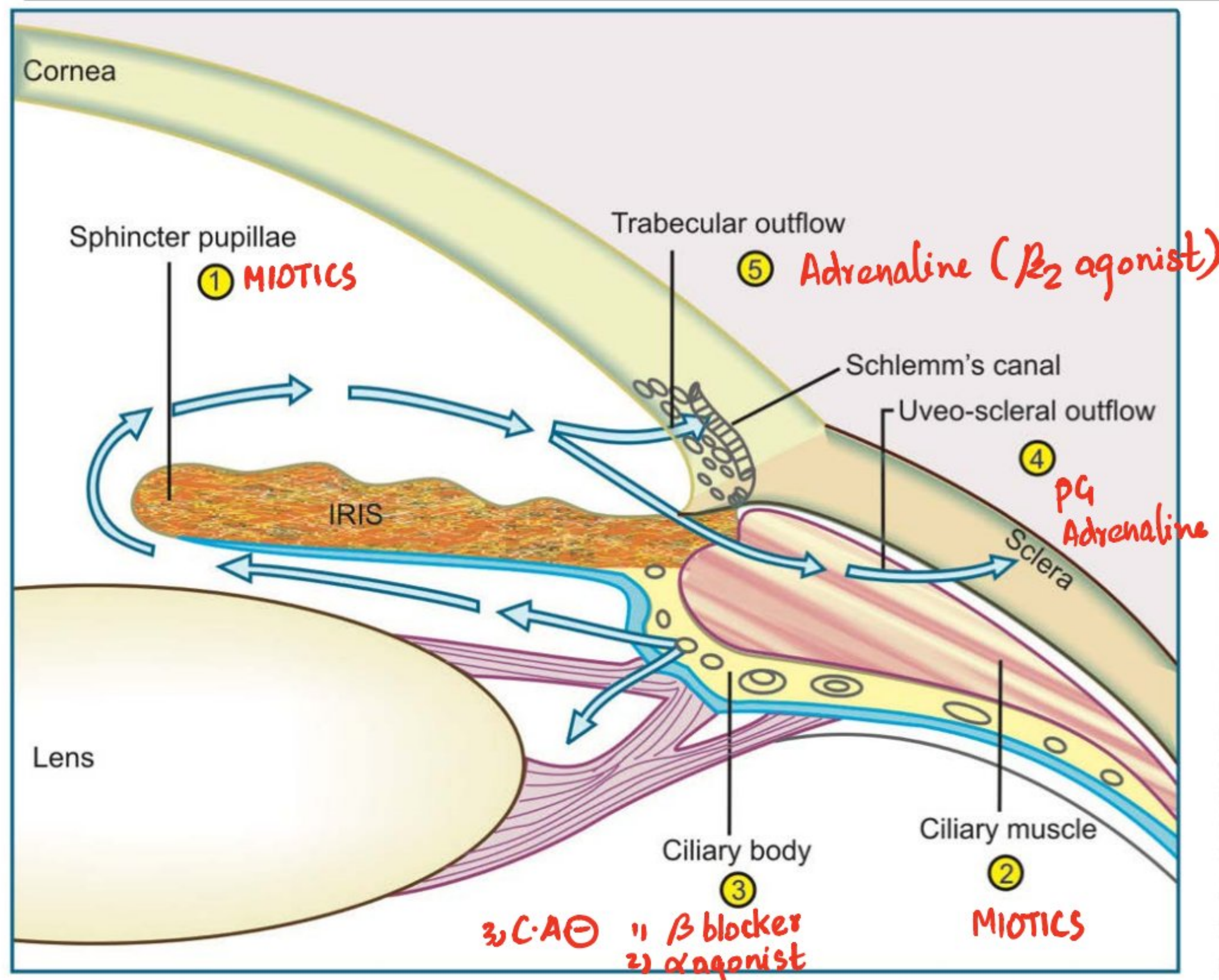


Fig. 10.1: Illustration of aqueous humor dynamics and the sites of action of ocular hypotensive drugs

- (1) Site of action of miotics in angle closure glaucoma: contraction of sphincter pupillae removes pupillary block and reverses obliteration of iridocorneal angle
- (2) Site of action of miotics in open angle glaucoma: contraction of ciliary muscle pulls on scleral spur and improves trabecular patency
- (3) Site of action of (a) β blockers (b) α₁ agonists (c) α₂ agonists (d) carbonic anhydrase inhibitors: all reduce aqueous secretion by ciliary body
- (4) Site of action of prostaglandins and adrenaline (α agonist action): increase uveoscleral outflow by altering permeability and/or pressure gradients
- (5) Site of action of adrenaline (β₂ agonist action): possibly increases aqueous conductivity of trabecular filtering cells

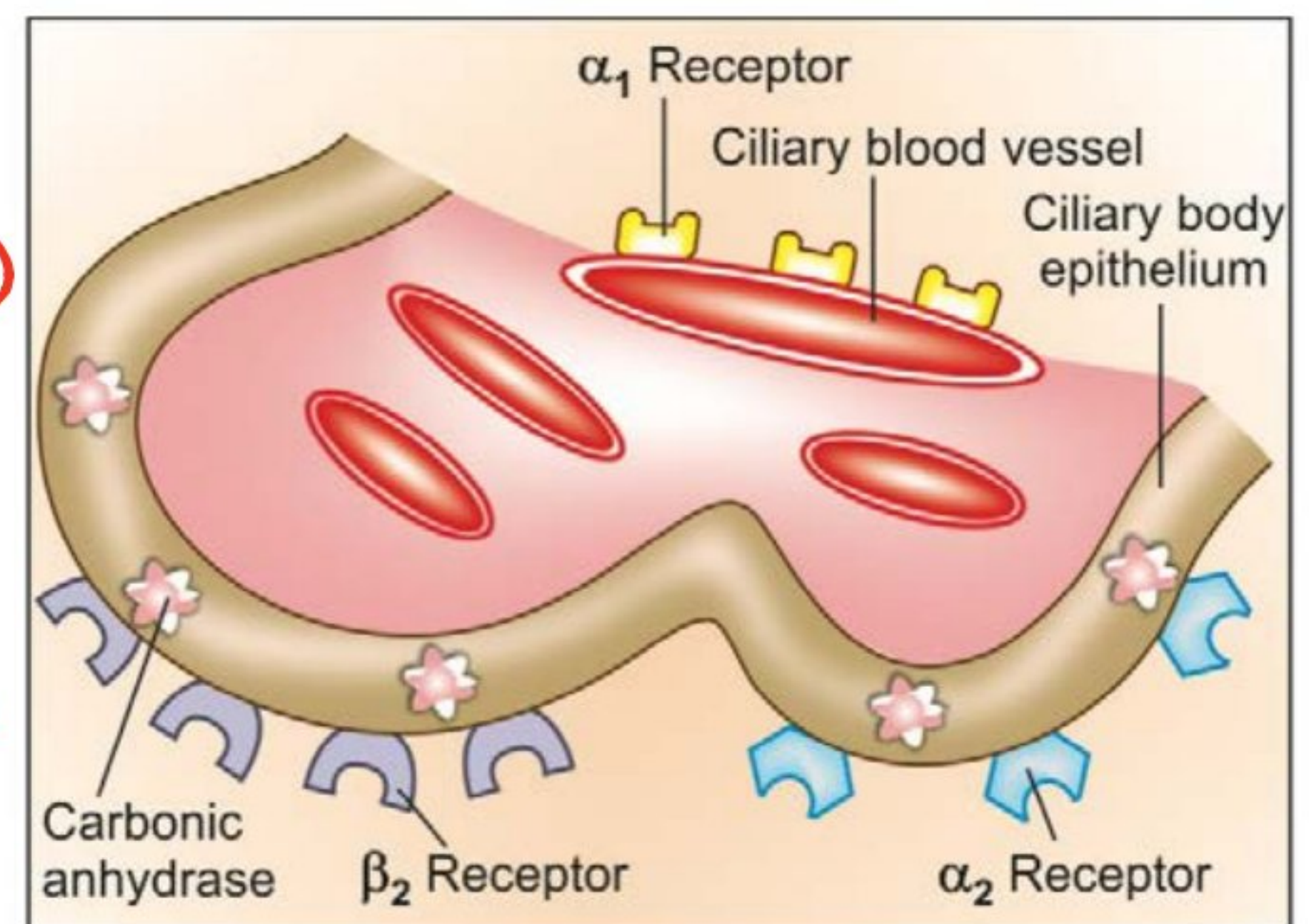


Fig. 10.2: Location of α₁, α₂ and β₂ adrenergic receptors and of carbonic anhydrase enzyme (molecules which are targets of drug action for reducing aqueous formation) in the processes of the ciliary body

α₁ adrenoceptors constrict ciliary vessels and reduce aqueous production: are activated by adrenaline.

α₂ adrenoceptors located on ciliary epithelium reduce aqueous secretion: are activated by adrenaline/apraclonidine/brimonidine.

β₂ adrenoceptors located on ciliary epithelium enhance aqueous secretion via increased cAMP. Their blockade by timolol/betaxolol reduces secretion.

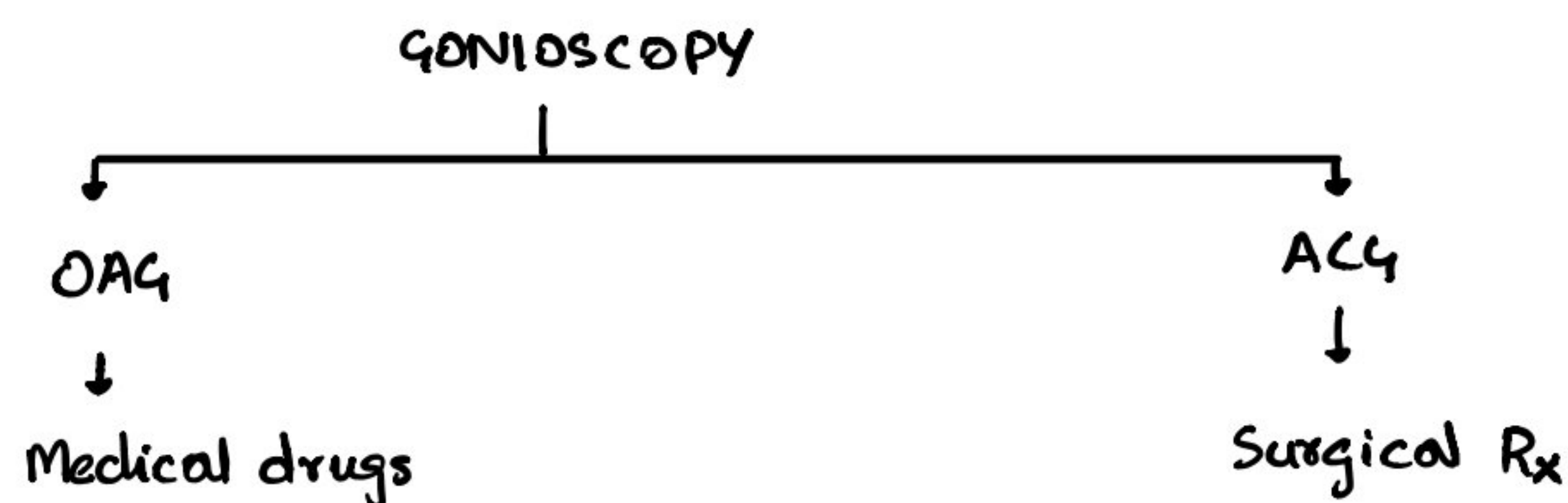
Carbonic anhydrase present within ciliary epithelial cells generates HCO₃⁻ ion which is secreted into aqueous humor. Acetazolamide/dorzolamide inhibit it and reduce aqueous secretion.

3. Prostaglandin analogues

Low concentration of $\text{PGF}_{2\alpha}$ was found to lower i.o.t without inducing ocular inflammation. It acts by increasing **uveoscleral outflow**, possibly by increasing permeability of tissues in ciliary muscle or by an action on episcleral vessels. An effect on trabecular outflow has also been demonstrated, but is less marked. Ciliary body COX-2 has been found to be down regulated in wide angle glaucoma indicating a physiological role of PGs in aqueous humor dynamics.

HOW TO PROCEED IN A SUSPECTED GLAUCOMA PATIENT

- First do Gonioscopy



LASERS FOR GLAUCOMA:

POAG: Laser Trabeculectomy → ↑ se outflow

ALT: Argon Laser

SLT: Selective laser (Diode laser)

↳ selectively absorbed by post-pigment TM

also used in :- 2° Glaucoma d/t pigment dispersion / pseudoexfol.

PACG → i, Peripheral iridotomy using Nd:YAG laser

• fired into iris crypts

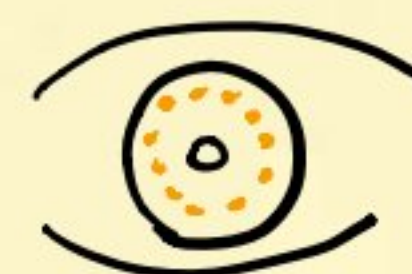
• ABRAHAM'S IRIDOTOMY LENS +66D



miotic used .. for better visualisation
thinning out iris

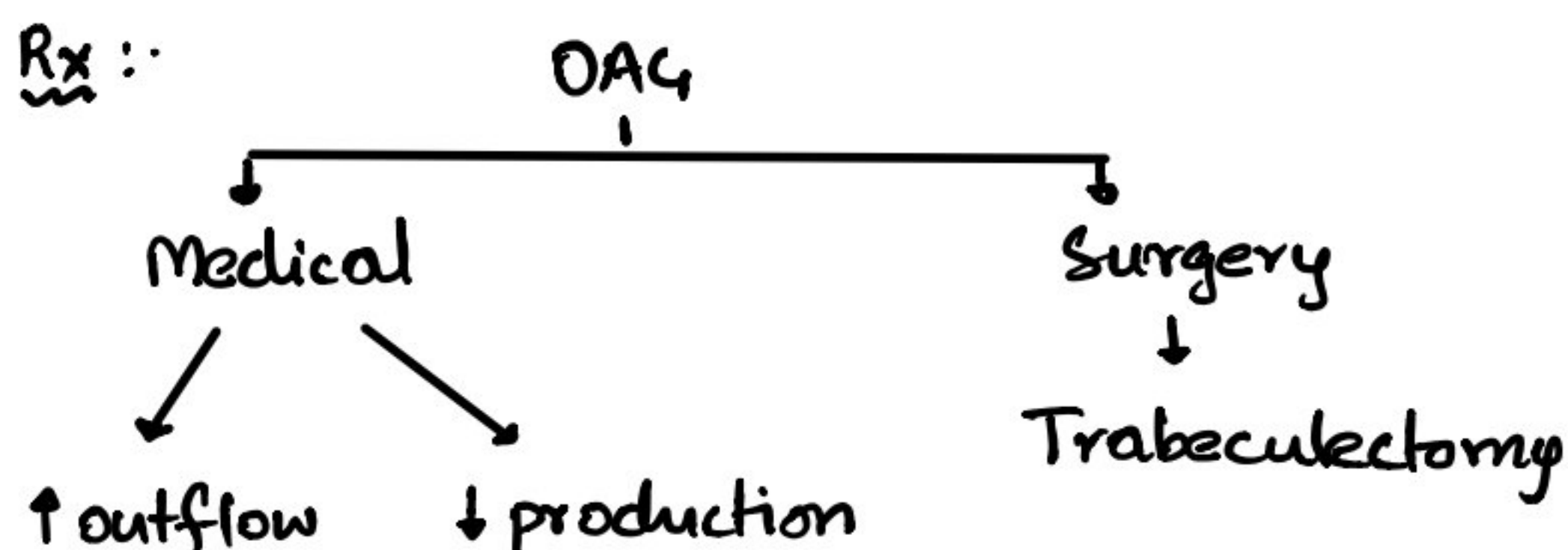
ii Laser Peripheral

iridoplasty → Argon laser



photocoagulation → no Iris Bombs formed

⇒ also useful in Pseudo iris



↑ outflow **Miosis?**

• Pilocarpine [M3 Agonist]

• Prostaglandin Analogue

↓ production

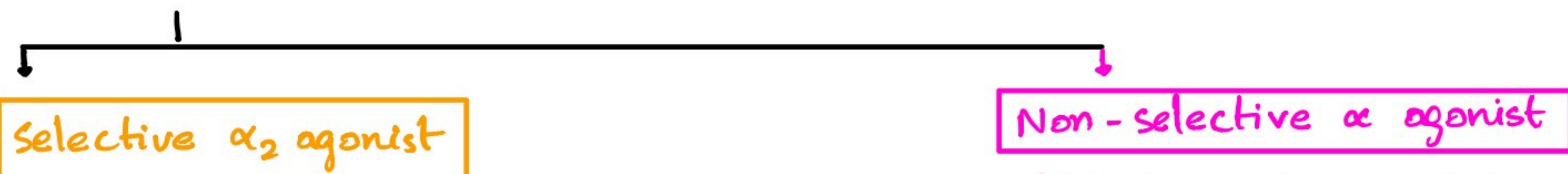
• α agonist

• β blockers

• Carbonic Anhydrase ⊖

① Drugs that ↓ IOP production:

α Agonist



Selective α_2 agonist

- C/I in children
- CNS suppression $\rightarrow$ sleep apnea
 - 1) Apraclonidine
 - 2) Brimonidine
 - 3) Clonidine

Brimonidine:

S/e: Endothelial decompression

Adv: • Neuroprotectant (prevent RGC death)

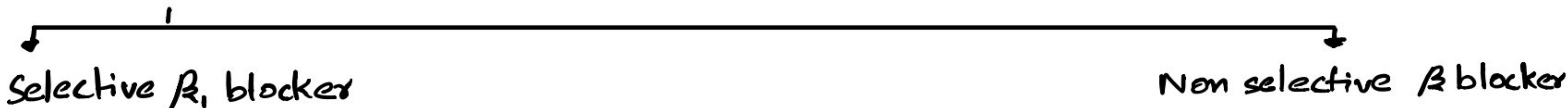
- ↑ uveoscleral outflow

Dipivefrine: • (MCC) of allergic blepharoconjunctivitis

Epinephrine: • Blackish discolouration of conjunctiva

- Adrenochrome deposits
- ARGYROSIS (Ag $\rightarrow$ silver)
- Cystoid macular edema in aphakic [C/I]

② β Blocker



Selective β_1 blocker

BETAXOLOL

- β blocker of choice in asthmatics
- ↑ blood flow to O.N
- Neuroprotective

Non selective β blocker

TIMOLOL

- C/I:
- i) Asthma
 - ii) AV block, Brady cardia
 - iii) CHF
 - iv) DM (Hypoglycemia)

3) Carbonic Anhydrase $\ominus$ $\rightarrow$ Sulfonamide



Systemic

- Acetazolamide

Topical

- Brimonamide
- Dorzolamide

- Side effects:
- i) A - Allergies (sulfa)
 - Metabolic Acidosis (sickle cell d/s, renal failure)
 - Aplastic anemia
 - ii) B - Bitter taste
 - iii) C - Calculi (renal)
 - iv) D - Dyspepsia
 - v) G - 2° ACU d/t anterior rotation of CB
 - vi) H - Hypokalemia
 - vii) P - Paresthesia (perioral & extremities)

II) Increase Aqueous Outflow:

- 1) ↑ trabecular outflow :: **PILOCARPINE**
 - parasympathomimetic
 - sphincter muscle → miosis → Rx of A/c attack
 - ciliary muscle
- Ciliary muscle (longitudinal muscle) contract → ↑ Trabecular outflow
- DOC: in A/c congestive glaucoma

Side effects of Pilocarpine

- 1) Lacrimal punctum stenosis
- 2) Iris Cyst
- 3) Cataract → Ant. polar cataract > ASC
- 4) Accommodation spasm → Induced myopia & painful blurring of vision
- 5) C/I in young myopes
- 6) C/I in inflammation → Accentuates Inflammation

Drugs ↑ uveoscleral outflow

PG Analogue = PGF_{2α}

- ↑ MMP (matrix metalloproteinases) → ↑ outflow to suprachoroidal space
- Independent of IOP

↓
uveoscleral outflow ↑

Eg:

- 1) Latanoprost
- 2) Travoprost
- 3) Bimatoprost
- 4) Unoprostone

PG Analogues

DOC ⇒ PAGA
 &
 PACG

↑ se both outflow

s/e of PG analogue

- 1) Trichomegaly
 - 2) Hyperpigmentation of Iris
 - 3) Pericardial hyperpigmentation
- ↳ C/I in Inflammation



III HYPEROSMOTICS ⇒ by ↓ se water content

- 1) IV Mannitol → C/I in CHF
CKD
- 2) Oral glycerol syrup → C/I in DM
§ glycerol metabolised to glucose §
- 3) Isosorbide → safe in DM
- 4) Urea 50% (IV) : old.

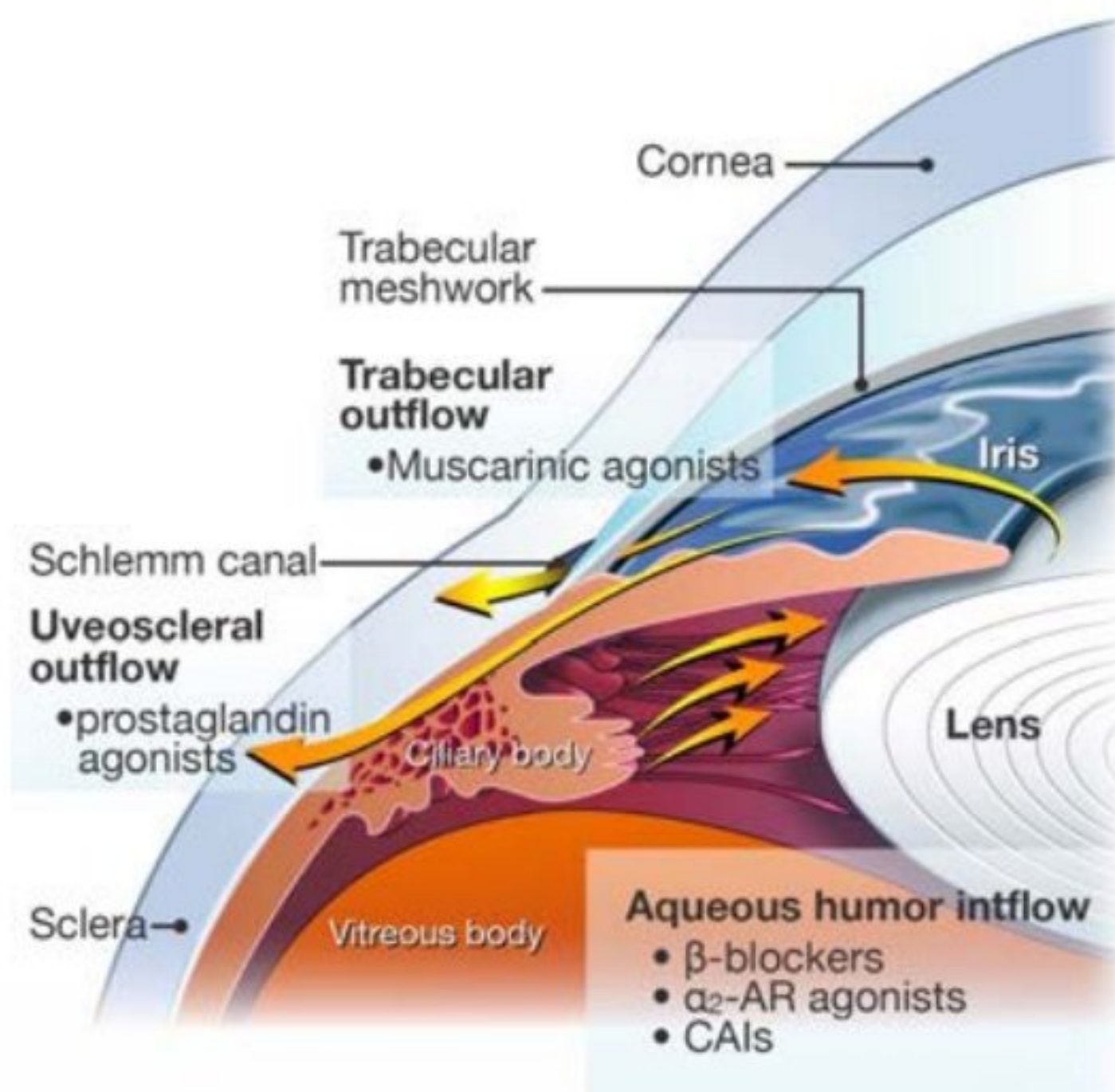
Other drugs ↑ se uveoscleral outflow {added fnx}

- 1) PG F_{2α} analogues
- 2) Non selective α agonist → • Dipivephrine
• Epinephrine
- 3) Brimonidine → α₂ agonist
- 4) Pilocarpine

Drugs ↑ se Trabecular outflow

- 1) Pilocarpin
- 2) Bimatoprost

⚠ DOX in HTN uveitis → Dipivefrine
§ ↑ se IOP §



FIRST LINE AGENTS OF OAG

- 1) PG analogue → Bimatoprost
- 2) Topical β blockers → Timolol
- 3) Selective α₂ agonist → Brimonidine

Adjuvant therapy

CA⊖

Secondary Glaucoma

• Neovascular

Retina (O_2 consumption $\uparrow$)



Hypoxia



VEGF (Vascular endothelial growth factor)



Neovascularization

(1st in retina $\rightarrow$ vitreous $\rightarrow$ lens and iris
(neovascularis iris)

- Rubeosis Iridis

- contraction of iris } Pulls the iris close to cornea $\rightarrow$
 $\uparrow$ IOP } (2^o ACG)

causes ::

- 1) Diabetes (MC)
 - 2) CRVO (Central retinal venous occlusion)
 - 3) Ocular ischemic disease
- Zipper like closure

Early ::

- $\otimes$ neovascularization [$\otimes$ VEGF or treat hypoxia]

Block VEGF [Anti VEGF drugs]

- 1) Bevacizumab } via pars plana.
- 2) Ranibizumab }

Pan-retinal photocoagulation $\rightarrow$ treatment of retinal hypoxia

$\downarrow$ IOP : Except latanoprost and pilocarpine (C/I)

Trabeculectomy

Late ::

IOP rises to 70-80mm $\rightarrow$ Glaucomatous optic atrophy



Blind

(absolute glaucoma: whenever patient becomes blind
d/t glaucoma)

Patient requests to $\downarrow$ pain : Cyclodestruction (ciliary body destruction)

- DLEP [Diode laser cyclo photo coagulation]
- Cyclocryopexy = -80°C probe

Surgical Mx of Glaucoma

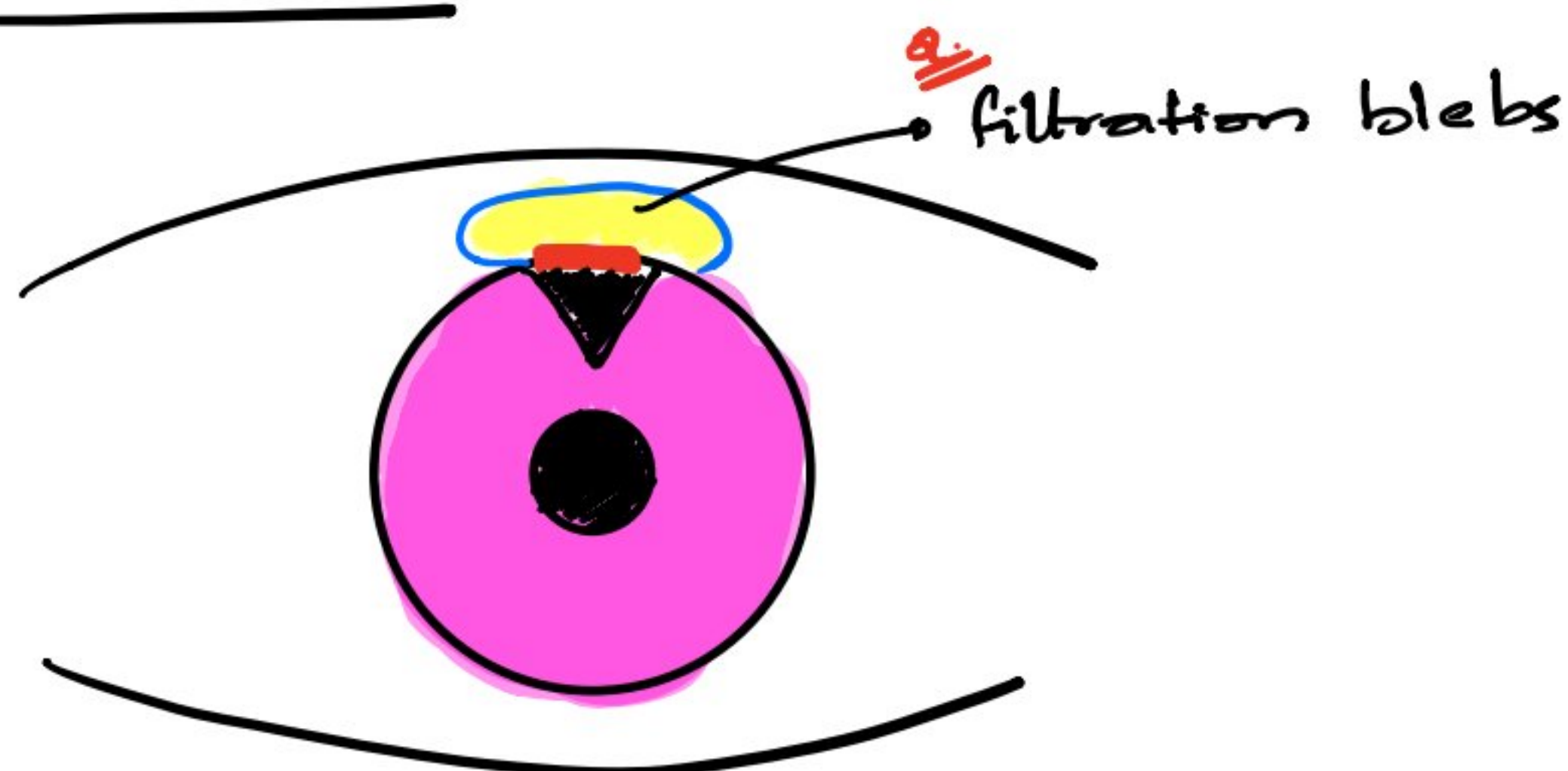
Medical Mx with 3 drug, still IOP not controlled



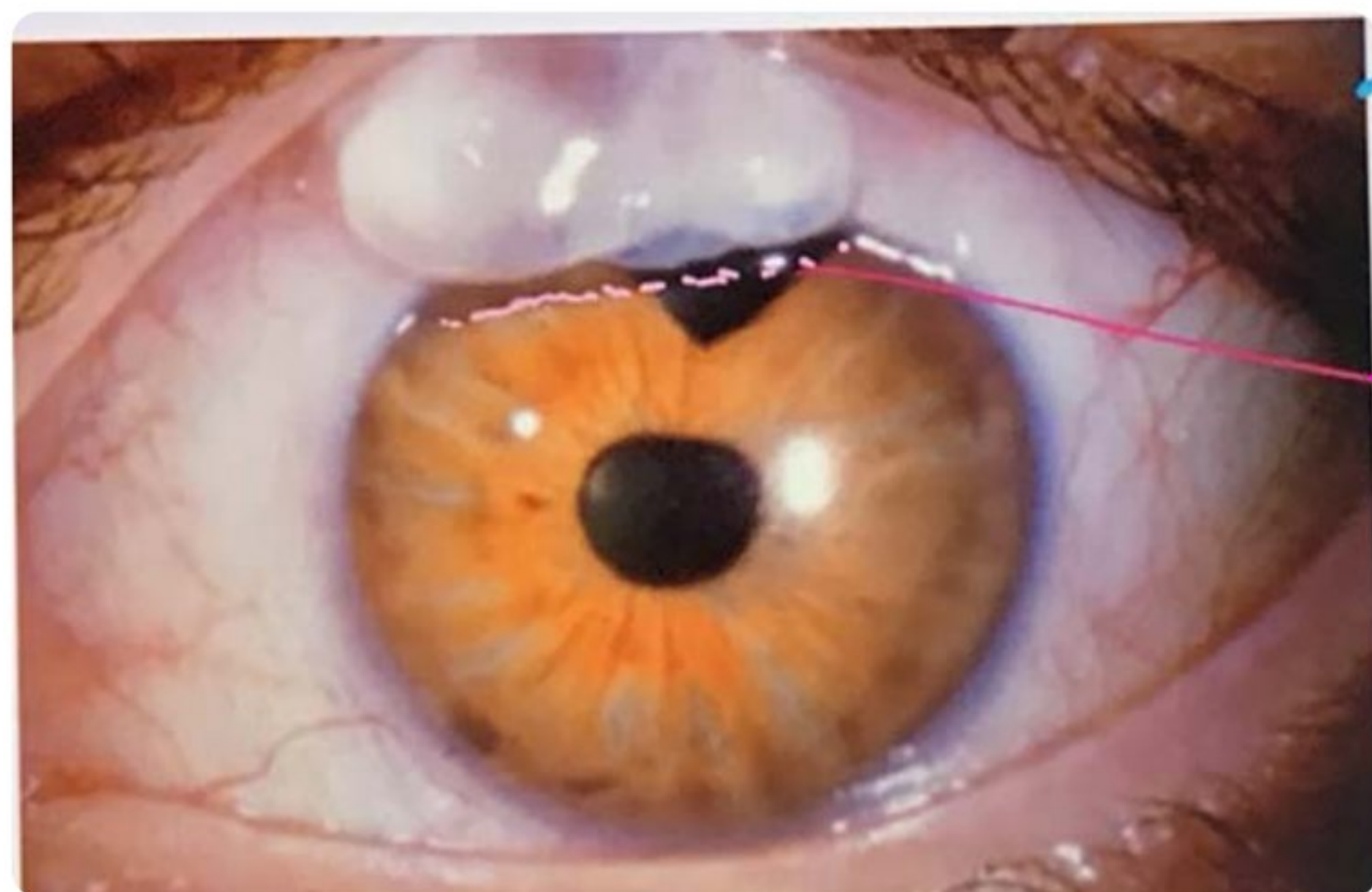
Go for SURGICAL Mx

- Gold STD Sx for POAG } **TRABECULECTOMY**
- PACG }

TRABECULECTOMY:



Trabeculectomy + peripheral superior iridectomy



→ Ferry's line : Fe^{2+} deposition @ Epithelium of filtration blebs

→ Aqueous drain into subconjunctival space

- Intraop i) Mitomycin C → antifibrotic agent @ scleral bed to prevent fibrosis of wound
 - ii) 5FU
 - iii) **OLOGEN** → **Biocompatible collagen**
- NOTE: AUGMENTED TRABECULECTOMY

s/e is overfiltration → Hypotonic Maculopathy
IOP $< 6\text{ mm Hg}$

2) Scleral necrosis → Staphyloma / scleromalacia perforans.

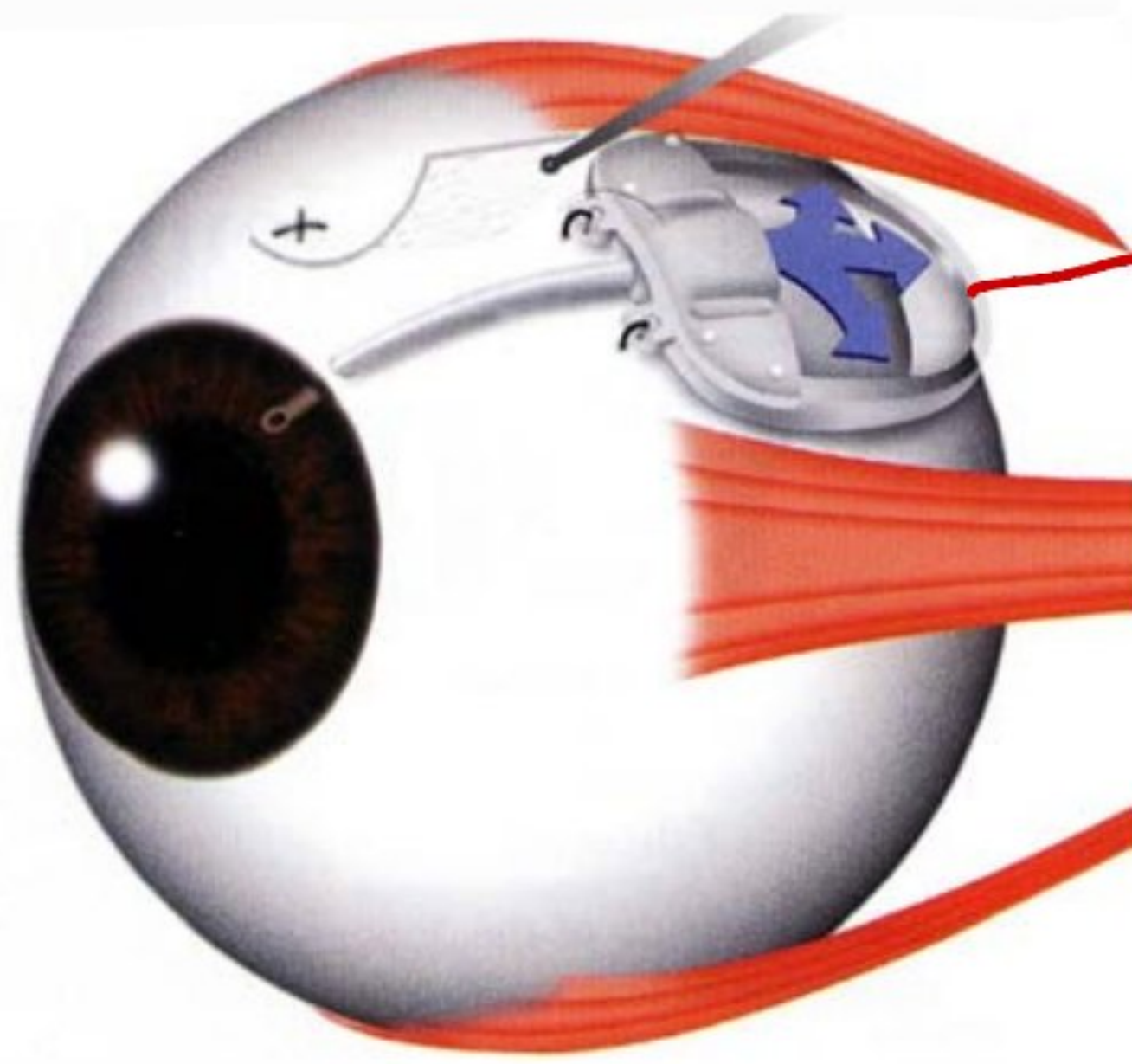
Failed Trabeculectomy



Aqueous Shunt / Glaucoma Drainage Device (Seton)

SETON

1 Ahmed Glaucoma Valve (AGV) → (MC) controlled filtration



2 Baerveldt implant

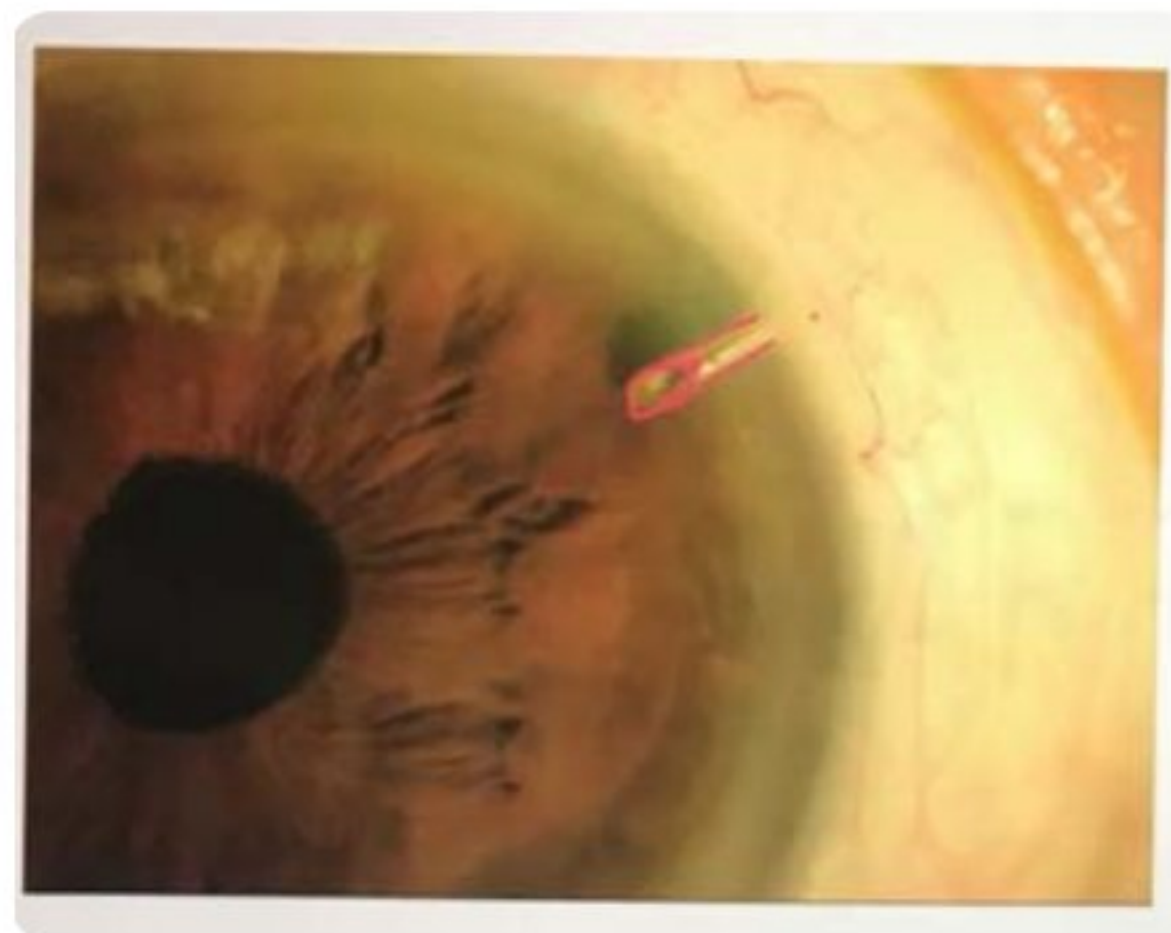
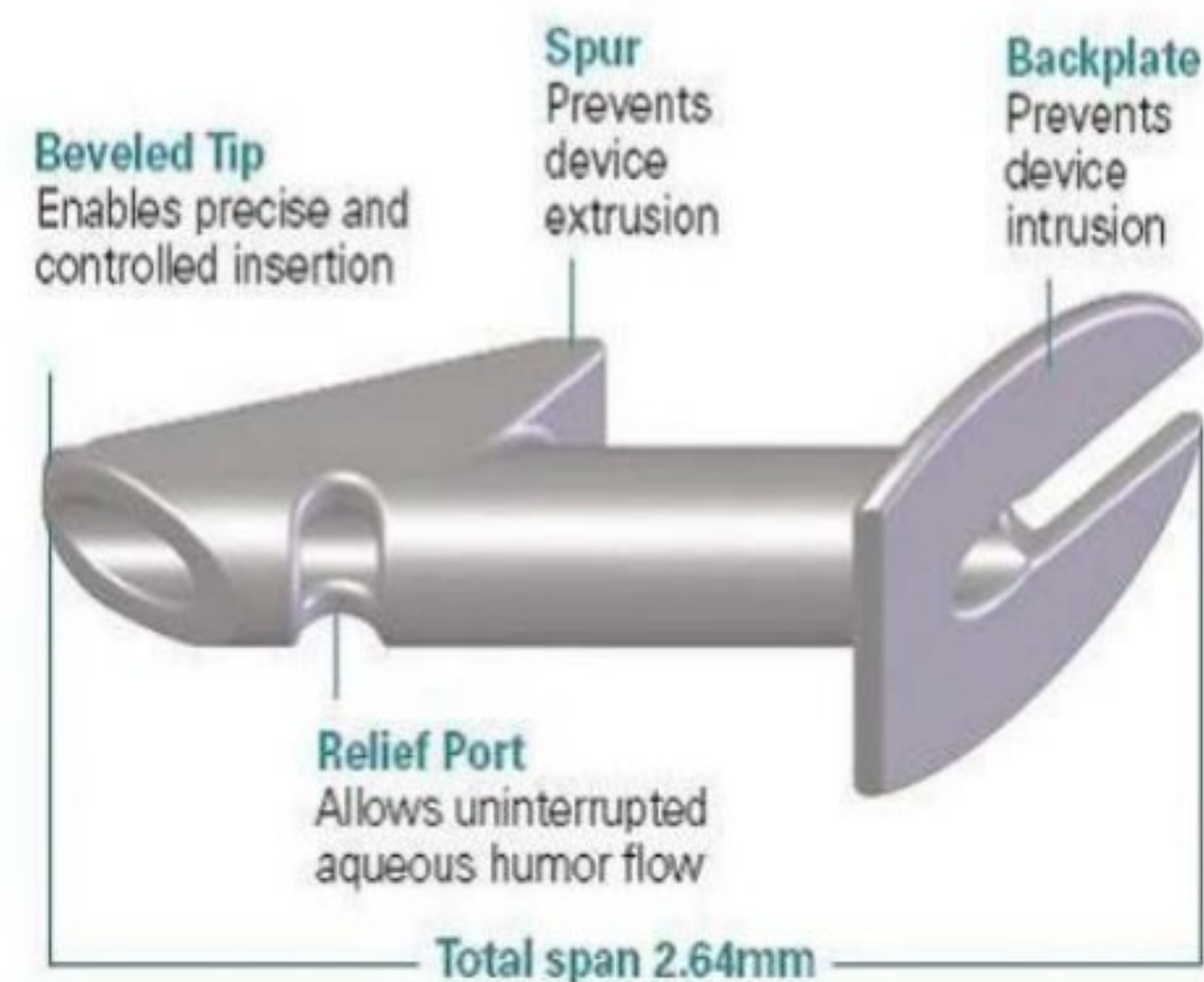
3 Identify the image



OTHER SHUNTS:

The Ex-PRESS™ Mini Glaucoma Shunt

→ made up of stainless steel



a) Express implant

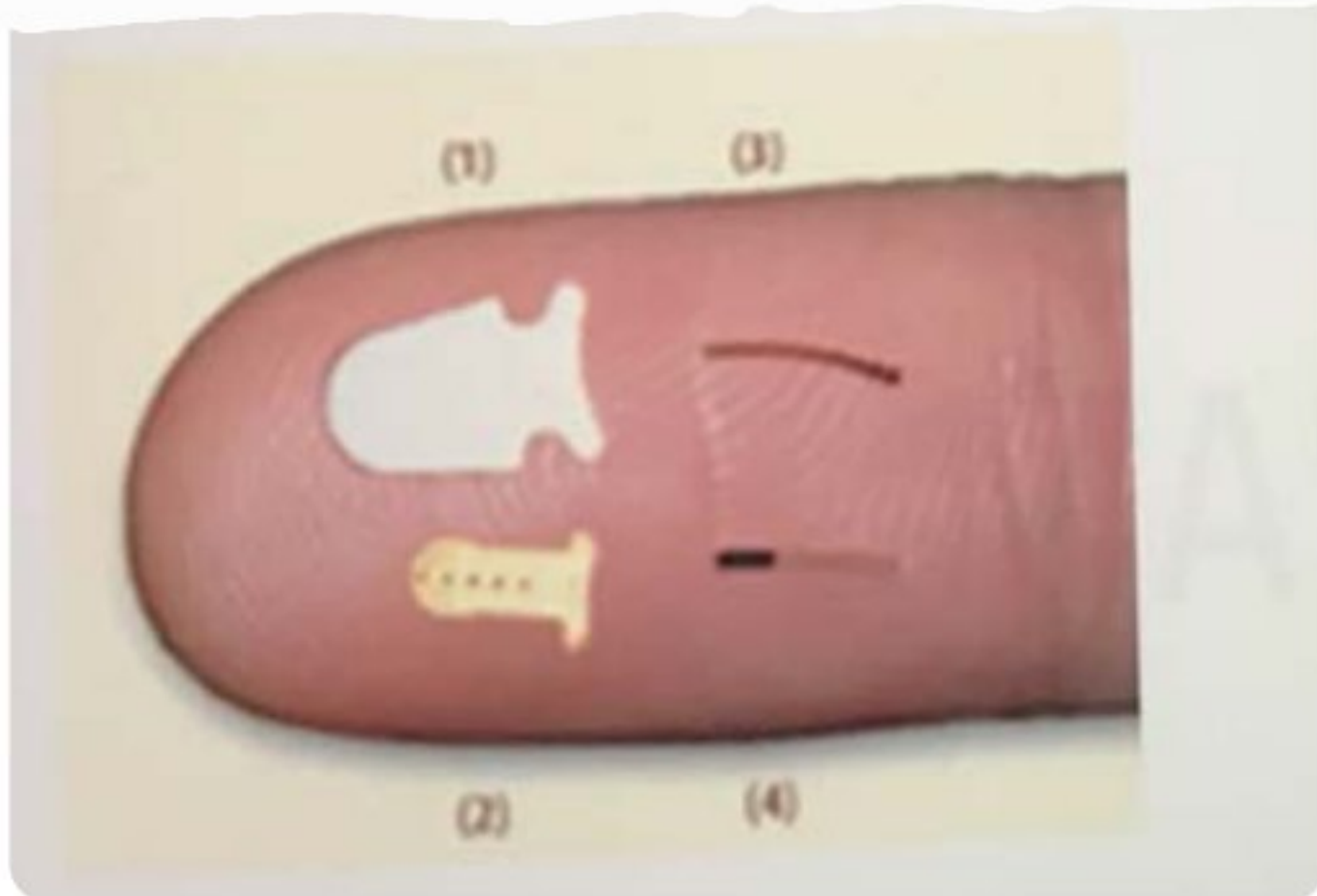
b) Gonioscope

✓ Baerveldt implant

d) Ozurdex implant

SUPRACHOROIDAL STENTS

→ Push aqueous to suprachoroidal space
• SOLX shunt → made of GOLD
• Hydra shunt



TRIPLE PROCEDURE OF GLAUCOMA:

- ① Trabeculectomy
- ② ECCE
- ③ PC-IOL implantation

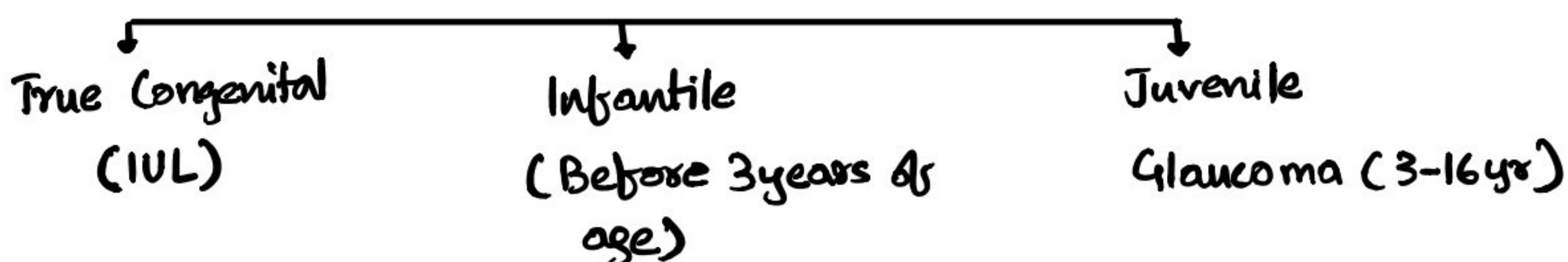
Developmental Glaucoma

- 1) Congenital glaucoma → Dx @ birth
- 2) Infantile glaucoma → upto 3yr of age
- 3) Juvenile glaucoma → 3-13yr of age

DEVELOPMENTAL GLAUCOMA

• Sporadic 90% , AR: 10%.

A) Primary congenital Glaucoma (most common form of Congenital Glaucoma)



GLC3 defect:

C/F : a) Triad of i) Photophobia ii) Blepharospasm iii) Epiphora

Corneal signs:

i) Corneal edema (cloudy cornea at birth)

ii) Corneal enlargement ($>13\text{mm}$)

$>11\text{mm}$ at birth megalocornea, $>12\text{mm}$ at 1yr of age to 2, $>13\text{mm}$ at any age.

• **Haab's striae** (Horizontal) or concentric to limbus $\rightarrow$ break in DM.

q **Vertical breaks $\rightarrow$ Trauma**

• **Buphthalmos** :- i) Blue and thin sclera, stretched limbus

ii) Iridodonesis

iii) Deep AC

iv) Axial myopia.

• **Glaucomatous cupping** $\rightarrow$ q **Reversible**

• IOP $\rightarrow$ Raised, but neither marked nor acute.

Differential Diagnosis:

a) Cloudy Cornea at birth (mnemonic STUMPED)

i) Birth trauma

ii) Rubella keratitis

iii) Metabolic disorders - mucopolysaccharidosis, lipidosis and cystinosis

b) Large cornea

i) Megalocornea

ii) Myopia

c) Lacrimation $\rightarrow$ Non patent NLD

d) Secondary Glaucoma

i) Retinoblastoma

ii) Juvenile Xanthogranuloma

iii) Persistent hyperplastic primary vitreous

iv) ROP

e) Rx : **Brimonidine Contraindicated**

i) Goniotomy if cornea is clear

ii) Trabeculotomy if cloudy cornea

iii) Combined trabeculotomy & trabeculotomy $\pm$ MMC (mitomycin) $\rightarrow$ TOC.

*** NOW A DAYS BEST MODALITY IS TRABECULOTOMY COMBINED with trabeculectomy +/- MMC.**

GLAUCOMA associated with congenital anomalies:

a) With irido - corneal dysgenesis

- i) Axenfield's anomaly = AR
- ii) Riegers anomaly
- iii) Riegers syndrome
- iv) Peters's anomaly = SPORADIC

b) Associated with ectopia lentis

- i) Marfan syndrome
- ii) Weil - Marchesani syndrome
(inverse glaucoma)

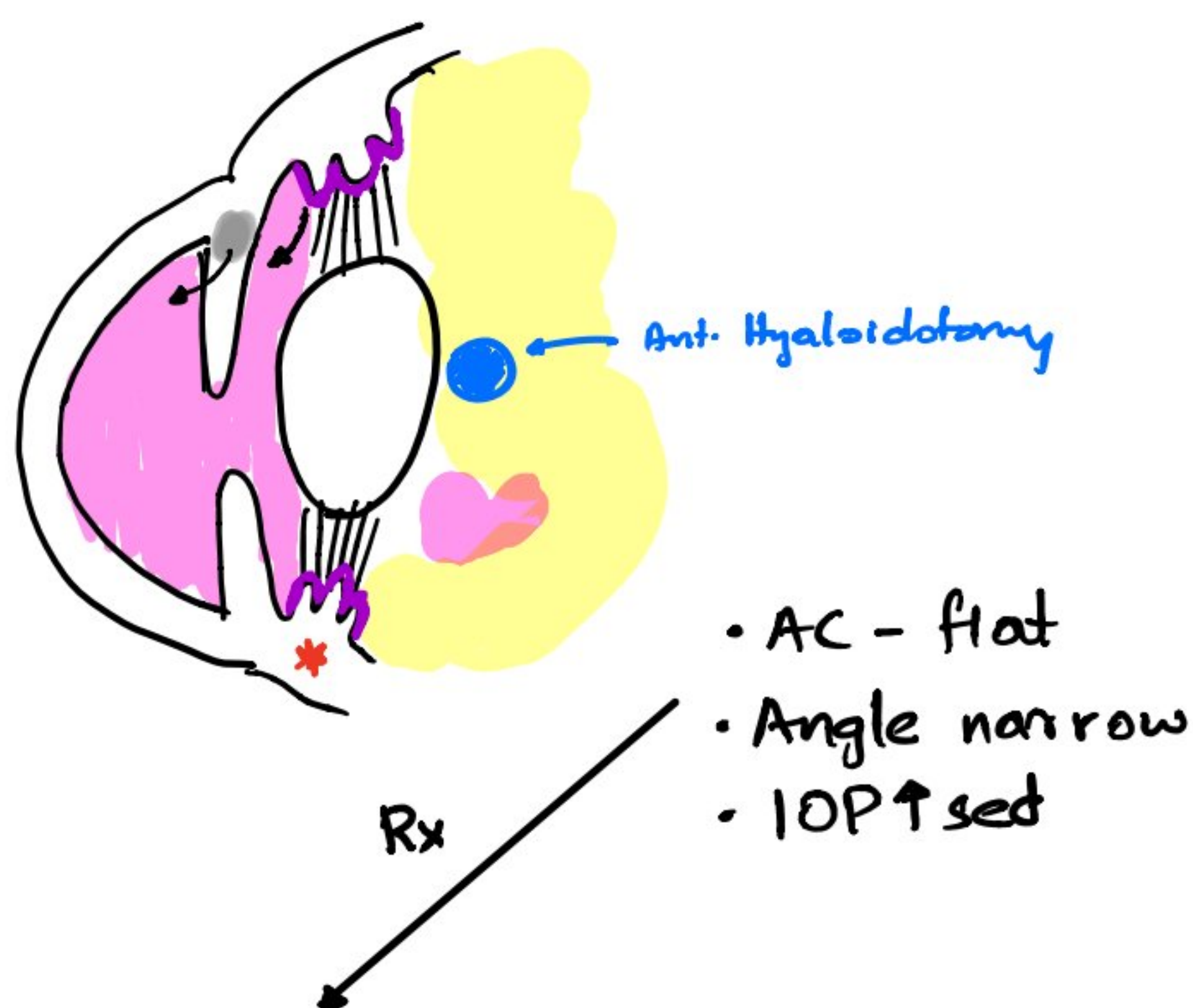
* 1) Post embryotoxon + anterior iris strands = Axenfield

2) Axenfield + iris hypoplasia = Riegers anomaly

MALIGNANT GLAUCOMA / INVERSE GLAUCOMA

a.k.a aqueous misdirection sx / ciliovitreal block

cause: After trabeculectomy → aqueous goes posteriorly & forms pockets in vitreous



↓
Vitreous IOP ↑ se
↓
Vitreous Push lens

↓
← { • Iris pushed forward
• Ciliary body rotated anteriorly

No response to laser peripheral iridotomy

↓ Rx

if not relieved → CYCLOPEGICS: ATROPINE } DOC for malignant G.I
Relieve the anterior rotation of ciliary bodies
Nd: YAG laser
Anterior Hyaloidotomy.

PIGMENT DISPERSION GLAUCOMA:

- Myopic males
- 2° OAG
- Melanin deposited on endothelium → KRUKENBERG'S SPINDLE
- " " on TM @ Schwalbe's line → SAMPOELESSI'S LINE

Rx: Selective laser trabeculectomy